DISCUSSION PAPER SERIES

DP21479

EXTREME WEATHER EVENTS, FISCAL SPACE, AND THE TERM STRUCTURE OF SOVEREIGN DEBT

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CLIMATE CHANGE AND THE ENVIRONMENT

ISSN 0265-8003

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Discussion Paper DP21479
Published 13 May 2026
Submitted 01 May 2026

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JEL Classification: Q54, E62, E52

Keywords: Climate change impacts

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Acknowledgements
We acknowledge financial support from the South African Reserve Bank (SARB) for this project. An earlier version titled "The Impact of Extreme Weather Events on the Term Structure of Sovereign Debt" was published in the SARB Working Paper series. We thank Tobias Adrian and colleagues at the International Monetary Fund for providing us with international panel data for sovereign debt yields and their components. We thank Joe Feyertag (discussant) and seminar participants at the SARB for helpful comments.

Extreme Weather Events, Fiscal Space, and the Term Structure of Sovereign Debt*

Emanuel Moench† Robin Schaal‡

April 26, 2026

Abstract

We study the impact of extreme weather events on global sovereign bond yields. Using state-dependent panel local projections, we show that the sovereign's fiscal position determines the sign of the yield response to natural disaster shocks. Yields decline significantly in fiscally constrained economies and rise in unconstrained ones, a pattern that remains masked in unconditional specifications. The reason is that government expenditure increases in unconstrained economies and remains flat in constrained ones, generating divergent inflationary dynamics that transmit to the yield curve. The state-dependent transmission operates through different channels. In advanced economies, the yield response is driven by the expected path of policy rates, while the term premium plays a larger role in emerging markets. Our findings highlight fiscal space as a key determinant of how climate shocks propagate through sovereign debt markets.

Keywords: Climate change, extreme weather events, sovereign debt, monetary policy

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1 Motivation

The increasing frequency and severity of extreme weather events due to anthropogenic climate change (IPCC, 2021) pose growing challenges for both fiscal and monetary authorities. Governments must respond to increasingly frequent and costly natural disasters while financing the transition to a low-carbon economy. These pressures interact with existing fiscal constraints and prompt central banks to adjust their policy stance and investors to reprice debt. Hence, sovereign debt markets may be affected by extreme weather events through multiple channels. Yet, little is known about how these events impact the term structure of interest rates and which role the fiscal position of the government plays in the transmission.

Using a panel of 18 advanced economies and twelve emerging markets with daily sovereign yield data, we estimate state-dependent local projections following Jordà (2005) and Auerbach and Gorodnichenko (2012) to trace the dynamic effects of severe weather events on sovereign bond yields and their components. Applying the decomposition of Adrian et al. (2013), we separate the yield response into an expected short rate path, capturing market expectations of future monetary policy, and a term premium, capturing compensation for interest rate risk. Our central finding is that extreme weather events generate opposite-signed yield responses depending on the sovereign's fiscal position prior to the shock. Strikingly, yields decline in fiscally constrained economies and rise in unconstrained ones. Specifications that do not condition on fiscal space show no significant yield response, masking the economically large heterogeneity that our state-dependent approach reveals.

We provide direct evidence for the fiscal space mechanism. Using government expenditure data from the IMF, we show that fiscally unconstrained sovereigns increase spending significantly following extreme weather events while constrained ones do not. This fiscal divergence generates distinct inflationary dynamics, visible in core CPI for advanced economies and food prices for emerging markets, that transmit into the yield curve through the expected policy path and term premium components.

In advanced economies, the yield response is primarily driven by the expected short rate path. High-debt countries experience a significant decline in the expected rate path, consistent with markets anticipating monetary easing in the absence of fiscal stimulus and inflationary pressure. Low-debt countries see rising government expenditures accompanied by core inflationary pressure, a positive expected rate path and a modest increase in the term premium which likely reflects anticipated sovereign debt issuance and inflation risk. In emerging markets, the term premium plays a substantially larger role, particularly among highly rated economies which see increases in both the term premium and expected rate path in response to rising government expenditures and food price inflation.

Combined, our results demonstrate that fiscal capacity is a key determinant of how extreme weather events propagate through sovereign bond markets. Countries with fiscal space cushion the economic fallout through reconstruction spending but face inflationary pressures and rising yields; countries without fiscal space experience monetary accommodation and falling yields. These findings have direct implications for sovereign debt management, monetary-fiscal coordination in the face of extreme weather events, and the modeling of climate risk in sovereign bond pricing.

The remainder of the paper is structured as follows. Section 2 reviews the literature and outlines the transmission channels through which extreme weather events affect the term structure of interest rates. Section 3 describes the data and empirical methodology. Section 4 presents and discusses our main results. Section 5 concludes.

2 Transmission Channels & Literature Review

Our work contributes to the growing literature on the macroeconomic effects of physical climate risks. Like much of the existing research, we focus on extreme weather events that can generally be associated with anthropogenic climate change, but we do not attribute individual events to climate change.[1]

2.1 Transmission Mechanisms

Standard neoclassical growth theory suggests that natural disasters can affect the macroeconomy through two primary channels. First, disasters such as storms and floods destroy physical capital, reducing an economy's productive capacity. This capital destruction initially lowers output, but subsequent investment may restore capital to its pre-shock level and bring the economy back to its steady state. Second, disasters such as droughts and heatwaves do not destroy capital but instead reduce productivity, effectively acting as a negative total factor productivity (TFP) shock. In this case, output remains below potential until productivity recovers.[2]

[1] The literature on attribution science establishes links between climate change and extreme weather events. Notable contributions include Min et al. (2011), Trenberth et al. (2015), and Stott et al. (2016).

[2] See Hallegatte and Przyluski (2010) for a broader discussion of how extreme weather events affect economic activity.

Empirical results by Felbermayr and Gröschl (2014) or Fomby et al. (2013), among others, confirm these predictions. Cantelmo (2022) argues that disasters reducing capital stock tend to be followed by an investment-driven recovery, whereas those that represent productivity shocks result in more prolonged economic contractions. These dynamics also have important implications for inflation. When disasters reduce supply by destroying capital, they can increase inflation by raising the marginal productivity of capital while compressing output. However, if extreme weather events increase uncertainty about future productivity and capital depreciation, risk-averse agents may shift toward precautionary savings, reducing investment and lowering aggregate demand. The net impact on inflation and output may vary based on whether supply- or demand-side effects dominate.[3]

2.2 Climate Shocks and Macroeconomic Outcomes

The literature on the economic impact of climate change can be broadly classified into two strands. The first examines the effect of extreme weather events on economic growth. Several studies (Noy, 2009, Dell et al., 2012, Burke and Tanutama, 2019, Acevedo et al., 2020, Kahn et al., 2019, Kiley, 2021, Kim et al., 2021) demonstrate that extreme weather events have negative effects on output growth, increasing the likelihood of economic contractions. Our work is particularly motivated by Fomby et al. (2013) and Felbermayr and Gröschl (2014), who not only document overall negative effects on growth but also highlight differences between advanced and emerging economies, as well as variations based on disaster magnitude. Although measured differently, both studies find stronger effects in developing economies and significant contractions in advanced economies. Related to that, Noy and Nualsri (2011) document counter-cyclical fiscal policy after disasters in developed economies and pro-cyclical responses in developing ones.

A second strand of research explores the relationship between extreme weather events and inflation. Beirne et al. (2022) find a significant positive link between natural disasters and headline inflation in the euro area, with heterogeneous responses across countries. Similar findings emerge for African economies in Kunawotor et al. (2022) and specifically for heatwaves in Faccia et al. (2021). Our study is particularly related to Parker (2018) who examines inflationary responses to natural disasters, distinguishing between country groups and disaster magnitudes. He shows that severe disasters lead to stronger inflationary pressures particularly in high-income countries. Chavleishvili and Moench (2025) document that natural disasters strongly shift the conditional distribution of output growth and inflation.

[3]Cantelmo (2022) builds on seminal work on disaster-induced macroeconomic shocks, including Barro (2006), Gabaix (2011), Gabaix (2012), and Gourio (2012).

They trigger an initial sharp increase of downside risk for growth, followed by a temporary rebound. They also lead to a temporary increase of upside risk to inflation. Ehlers et al. (2025) analyze the effect of extreme weather events on GDP and inflation as well as their subcomponents at the aggregate and sectoral level in a global panel. They find a persistently negative effect on output, but a relatively short-lived effect on inflation, driven in particular by food and energy prices. For a more comprehensive review of the rapidly growing literature on the impact of natural disasters on economic outcomes, see Tol (2018) or Botzen et al. (2019).

While these studies do not assess the impact of natural disasters on government debt markets, they suggest that investors are likely to incorporate macroeconomic risks from extreme weather events into sovereign debt pricing, particularly through their effects on growth, inflation, and the resulting response of monetary and fiscal policy.

2.3 Climate Risk and Sovereign Debt Markets

Recent research has examined how climate risks influence sovereign bond markets. Cevik and Jalles (2022) show that climate vulnerability significantly increases bond yields and spreads, particularly for developing countries. Similar effects on spreads and sovereign default premiums are documented by Klomp (2015) and Kling et al. (2025), while Painter (2020) finds comparable patterns for U.S. municipal bonds. Lis and Nickel (2010) estimate that budgetary costs from extreme weather events range from 0.23% to 1.4% of GDP in developing economies. Using a panel of 115 countries from 1985–2010, Klomp (2017) finds that large-scale natural disasters increase the probability of sovereign default by about three percentage points. Relatedly, Noy and Nualsri (2011) document counter-cyclical fiscal responses to disasters in developed economies and pro-cyclical responses in developing ones. Mallucci (2022) shows that in the Caribbean, hurricane risk reduces governments' ability to issue debt, with adverse welfare effects. In addition, Bauer and Rudebusch (2023) argue that declines in the natural rate of interest amplify the social cost of carbon, supporting stronger climate mitigation policies. Closely related to our work, Anyfantaki et al. (2025) estimate impulse responses of 10-year headline government yields at an annual frequency and distinguish between high and low debt states. We build on their insights by studying the role of fiscal states and impulse responses of yield components in detail, and relating it to the state-dependent response of inflation to extreme weather events. For a more comprehensive literature review on climate change and government debt, see Seghini (2024).

Despite this growing body of research, the understanding of the precise channels through which extreme weather events affect government bond yields remains underexplored. While the above mentioned studies establish that climate risks affect sovereign debt pricing, none decompose yield responses into their monetary policy expectation and risk premium components. In doing so, we shed additional light on the term structure response by studying how the expected path of policy rates and the term premium react to extreme weather events conditional on the fiscal state of the economy. Moreover, we document that the state-dependent response of yields to extreme weather events is consistent with that of government expenditures and inflation.

2.4 Yield Curve Decomposition and Extreme Weather Events

To investigate how extreme weather events affect sovereign bond yields, we rely on the model of Adrian et al. (2013), which expresses local currency bond yields as the sum of expectations about future short-term interest rates and a term premium. Specifically, for a local currency bond of maturity n, we decompose the observed yield $y_t^{(n)}$ as follows:

$$y_t^{(n)} = \tilde{y}_t^{(n)} + tp_t^{(n)}. \tag{1}$$

Here, $\tilde{y}_t^{(n)}$ represents the risk-neutral yield component (i.e., the expected average future short rate over the bond's life, abstracting from convexity) and $tp_t^{(n)}$ captures the term premium, which reflects compensation for interest rate risk. In structural models, the term premium is driven by uncertainty about inflation and real short rates, supply-and-demand dynamics in sovereign debt markets, and broader risk sentiment. Since we rely on local currency sovereign yields, the central bank retains the capacity to monetize government debt, precluding nominal default on domestic-currency obligations — though potentially at the cost of inflation. This allows us to decompose observed yields into an expected policy rate path and a term premium without requiring a separate sovereign credit risk component. Assuming that log bond prices are affine in their pricing factors, time-varying risk premiums can be estimated from market yields following the regression based approach established by Adrian et al. (2013).

Given the documented effects of extreme weather events on output, inflation, and macroeconomic uncertainty, climate shocks could affect the yield curve through two main channels:

1. *Monetary policy expectations:* If investors anticipate that central banks will respond to an extreme weather event by adjusting policy rates, this should be reflected in the expected short-rate path.

2. *Risk perceptions and term premiums:* If extreme weather events alter perceptions of inflation risks, fiscal sustainability, or economic uncertainty, they may influence the term premium.

By applying this decomposition, we aim to understand how investors perceive the macroeconomic consequences of climate shocks and whether their pricing of sovereign debt aligns with macroeconomic theory. We also investigate which state variables shape the cross-sectional and time-series variation in yield responses, with a particular focus on fiscal capacity. Importantly, since we rely on local currency yields, our setting does not focus on the direct credit risk concerns of international investors. To the best of our knowledge, ours is the first study to examine the immediate impact of extreme weather events on the term structure of sovereign yields and its underlying components.

3 Data & Empirical Approach

3.1 Data

Extreme Weather Events To identify extreme weather events and their magnitude, we rely on the EM-DAT database provided by the Centre for Research on the Epidemiology of Disasters (CRED) at the Université Catholique de Louvain. EM-DAT systematically records country-level human and economic losses from disasters that meet at least one of the following criteria: (i) at least 10 fatalities, (ii) at least 100 affected individuals, (iii) an official declaration of a state of emergency or (iv) a call for international assistance. This dataset is widely used in cross-disciplinary research on disaster impacts.

We focus exclusively on extreme weather events that can be linked to climate change, categorized as follows:

1. *Meteorological events:* storms, extreme temperatures, fog.

2. *Hydrological events:* floods and landslides.

3. *Climatological events:* droughts, glacial lake outbursts, wildfires.

Other natural disasters in EM-DAT – such as earthquakes, tsunamis, volcanic eruptions and epidemics are excluded because they are unrelated to climate change. EM-DAT compiles its data from a wide range of sources, including government agencies, United Nations bodies, NGOs, international aid organisations, insurance firms, research institutes and the press (CRED, 2025).

For our baseline analysis with daily data, we define the occurrence of an extreme weather event using an indicator variable equal to one on days when EM-DAT records a disaster in a given country. To distinguish between smaller and larger events, we measure the human impact as the sum of reported deaths, displaced individuals and those classified as affected by the event in the EM-DAT dataset. This is in line with much of the literature.[4] Following previous works studying the macroeconomic effects of extreme weather events, we focus on larger events. Specifically, we compute the distribution of human impact across all events of a given country in the EM-DAT database since its inception and then restrict our empirical analysis to the largest 25% of events per country across the entire database. From this subset we then consider the events that fall into our sample period as determined by the available yield data per country. This allows us to focus on the larger events per country while maintaining a sufficient panel across countries and time. That said, the absolute size of the impact varies strongly across countries.[5].

Identification Our identification relies on the assumption that extreme weather events in the top quartile of human impact are exogenous to contemporaneous financial market conditions. Two features of our setting support this assumption. First, while the *occurrence* of certain event types is partially predictable, hurricane seasons are known, flood-prone regions are mapped, the *severity* of any given event conditional on occurrence is essentially stochastic. Even professional forecasts cannot reliably predict whether a given event will produce top-quartile human impact until shortly before it materializes, well within the daily frequency of our analysis. Our shock variable thus captures the surprise component of disaster severity rather than the predictable component of disaster occurrence. Second, the severity of human impact, measured as the sum of deaths, injuries, displacement and affected individuals, is determined by the interaction of meteorological intensity with local geographic and demographic conditions in ways that are plausibly orthogonal to the sovereign's fiscal or financial market position at the time of the event. We verify this assumption empirically: Figure 14 in Appendix E presents event study plots spanning 10 days before to 20 days after the shock across all four fiscal regime–country group combinations. Pre-event yield movements are economically small and statistically insignificant, with no evidence of anticipatory movements in the direction of the post-event response.

[4]See for example Parker (2018), Fomby et al. (2013), Ehlers et al. (2025) or Anyfantaki et al. (2025) amongst many others.

[5]The ultimate count of disasters in a given country in our sample can be less than 25 percent of the total number depending on the availability of yield curve data for the respective country. In Appendix D we provide results with top 33% and top 10% as thresholds.

Yield Curve Data To examine the impact of extreme weather events on sovereign debt yields, we use local-currency zero-coupon yields sourced from Bloomberg. Specifically, we use a dataset provided by the International Monetary Fund (IMF), which applies the methodology of Adrian et al. (2013) to a large cross-country panel. This dataset has been employed in several studies, including Moench (2019) and Adrian et al. (2019), which analyse global bond yield co-movements, and more recently, Adrian et al. (2024), which examines the effects of U.S. monetary policy shocks on global sovereign bond yields and their components.[6]

The extracted expected rate path and term premium components exhibit economically plausible co-movements with macroeconomic fundamentals across both advanced and emerging market sovereigns. We note that the decomposition is likely more precisely estimated for advanced economies with deep and liquid government bond markets than for some developing economies, where thinner secondary markets may occasionally introduce measurement noise into the extracted components. This consideration reinforces our decision to present AE and EM results separately and to interpret EM estimates with appropriate caution regarding statistical precision. Importantly, since we rely on local currency yield curves the term premium does not include a default risk premium.

Our main sample consists of 18 advanced economies and twelve emerging markets. These countries feature developed financial markets and institutions but vary in their fiscal position and exposure to climate risks. An overview of all countries considered, data availability and the respective number of extreme weather events can be found in Table 2.

Macroeconomic and Financial Data We obtain additional macroeconomic and financial data from three primary sources. Exchange rates are obtained from Bloomberg. Macroeconomic data such as inflation rates, growth rates, debt ratios and government expenditures are obtained from the IMF Data Portal. CPI subcomponents are from Bloomberg. The fiscal position of advanced economies is measured by the net debt-to-gdp ratio. To study the fiscal position of emerging markets, we rely on the World Bank database for fiscal space based on Kose et al. (2022) and specifically use the foreign currency long-term sovereign debt ratings index.

3.2 Baseline Local Projections

Our empirical strategy is based on local projections, a flexible methodology introduced by Jordà (2005). This approach allows us to estimate impulse response functions without im-

[6] In contrast to the five-factor model proposed by Adrian et al. (2013) as their preferred specification, the international term premium decompositions use four principal components of yields as pricing factors.

posing strong parametric restrictions, making it well-suited to capturing the dynamic effects of extreme weather events on bond yields. Under mild assumptions, lshow that local projections and VARs estimate the same impulse responses, as demonstrated by Plagborg-Moeller and Wolf (2021). We estimate the following specification:

$$\Delta_{i,h} y_t^{(n)} = \alpha_i^{(n)} + \alpha_t^{(n)} + \beta_{1,h}^{(n)} \omega_{i,t} + \gamma_{i,h}^{(n)\prime} X_{i,t} + \varepsilon_{i,h,t}^{(n)} \tag{2}$$

where

$$\Delta_{i,h} y_{i,t}^{(n)} = y_{i,t+h}^{(n)} - y_{i,t-1}^{(n)}$$

is the h-period cumulative change of a yield (component) of a bond with maturity n for country i. In all regressions, dependent variables are winsorized at 1% and 99% to remove outliers. In these regressions, $\beta_{1,h}^{(n)}$ is the coefficient of interest measuring the cumulative yield (component) change following an extreme weather event.

In our baseline specification based on daily data, $\omega_{i,t}$ is a dummy variable equal to one if an extreme weather event occurred in country i on day t. In our monthly or annual regressions, $\omega_{i,t}$ measures the number of events occurring in country i in period t. As part of our vector of control variables X_i, we include three lags of the yield component ($y_{i,t-1}$, $y_{i,t-2}$, $y_{i,t-3}$) to account for potential serial correlations. We also control for the most recent annual reading of CPI inflation and GDP growth and country i's exchange rate with the U.S. dollar (and the euro in the case of the United States). Finally, we also include year fixed effects $\alpha_t^{(n)}$ to account for low-frequency shifts in global yields unrelated to natural disasters and country fixed effects $\alpha_i^{(n)}$ to capture country-specific structural differences. Given our panel regression setting, we compute Driscoll-Kraay standard errors (Driscoll and Kraay, 1998) to account for cross-sectional dependence.

3.3 State Dependent Impulse Responses

As discussed in Section 2, the economic consequences of extreme weather events may vary based on a country's fiscal position. Fiscally constrained economies might be less able to provide economic support following a extreme weather events or finance disaster relief by cutting down other expenditures, while countries with more fiscal space might increase debt issuance to implement stimulus measures. Hence, the fiscal position can be expected to shape the impact natural disasters have on countries' yield curves. To assess whether such state dependence is relevant empirically, we estimate state-dependent impulse responses following

Auerbach and Gorodnichenko (2012) and others.[7] Specifically, we modify Equation (2) to allow for a smooth transition between high- and low-debt regimes:

$$\Delta_{i,h} y_t^{(n)} = \alpha_i^{(n)} + \alpha_t^{(n)} + \beta_{1,h}^{(n)} F(z_{i,t-1})\omega_{i,t} + \beta_{2,h}^{(n)}(1 - F(z_{i,t-1}))\omega_{i,t} + \gamma_{i,h}^{(n)\prime} X_{i,t} + \varepsilon_{i,h,t}^{(n)}, \quad (3)$$

where $z_{i,t-1}$ is the state variable capturing fiscal conditions. For advanced economies this is measured as net public debt as a percentage of GDP and for emerging markets by the World Bank sovereign debt ratings index. The transition function is defined as

$$F(z_{t-1}) = \frac{exp(\theta \frac{z_{t-1}-c}{\sigma_z})}{1 + exp(\theta \frac{z_{t-1}-c}{\sigma_z})}, \quad (4)$$

where c and σ_z are the median and standard deviation of z_{t-1} across all countries in a given year. Debt-to-GDP is omitted from the control vector since $F(z_{i,t-1})$ is a monotonic transformation of the state variable, and the interaction terms already condition on the sovereign's fiscal position, making the inclusion of the level redundant. In line with Auerbach and Gorodnichenko (2012) we set $\theta = 1.5$ to ensure a smooth yet meaningful separation between fiscal regimes.[8]

Finally, to estimate the state-dependent impulse response functions of annual government expenditures in percent of GDP and monthly year-over-year consumer price inflation, we adjust Equation (3) to

$$\Delta_h q_{i,t} = \beta_{0,h} + \beta_{1,h} F(z_{i,t-1})\omega_{i,t} + \beta_{2,h}(1 - F(z_{i,t-1}))\omega_{i,t} + (n)\prime X_{i,t} + \varepsilon_{i,h,t}, \quad (5)$$

where $q_{i,t} \in \{\pi_{i,t}, g_{i,t}\}$.

4 Results

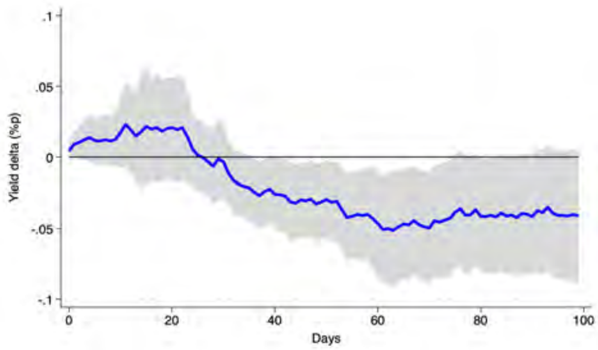
We begin by estimating unconditional impulse responses based on Equation (2) that pool all observations within each country group without conditioning on fiscal space. Figure 1 presents the response of the 2-year sovereign yield to an extreme weather event for advanced economies and emerging markets separately.[9] In both panels, the point estimate is economically small and mostly statistically indistinguishable from zero throughout the 100-day

[7] Works that have applied this approach to local projections include Tenreyro and Thwaites (2016) or Born et al. (2020).

[8] The estimated functions are shown in Appendix B. In Appendix B we also document robustness of our findings for values of $\theta = 3$ and $\theta = 5$.

[9] We present the same analysis for 10-year yields in Appendix C.

(A) 2Y yield: Advanced Economies



(B) 2Y yield: Emerging Markets

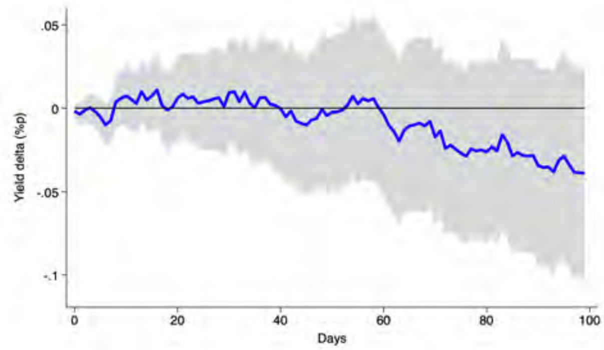


Figure 1: **Impulse responses of 2-year yields to extreme weather events (Daily).** The figure shows impulse response functions of the daily two-year yield to the largest 25% of extreme weather events, as specified in Equation (2). The left panel shows the impulse responses of countries in advanced economies and the right panel in emerging markets. The sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

horizon. The unconditional average effect of extreme weather events on sovereign yields is, to a first approximation, zero. As we show below, this null result masks economically large and statistically significant heterogeneity across fiscal regimes which becomes only visible once we condition on sovereigns' pre-existing fiscal positions.

4.1 Main Result

Table 1 reports the state-dependent impulse responses of the 2-year sovereign yield and its Adrian et al. (2013) decomposition into an expected short rate path and a term premium component, evaluated at the 50-day horizon, following Equation (3). The headline finding is a sharp and statistically significant sign switch across fiscal regimes. In fiscally constrained advanced economies, severe weather events reduce the 2-year yield by about ten basis points (significant at the 1% level), while in fiscally unconstrained advanced economies the response is positive at roughly six basis points, yielding a spread of more than 15 basis points associated with the sovereign's pre-existing fiscal position. In emerging markets, the pattern is directionally consistent but larger in magnitude: the 2-year yield declines by about 16 basis points in low-rated economies and rises by roughly 13 basis points in higher-rated economies, a significant spread of more than 28 basis points.

The decomposition of sovereign yields into expected policy path and term premium components reveals that this heterogeneity is not driven by a single channel, but instead reflects systematically different transmission mechanisms across country groups. In advanced economies, the expected short rate path accounts for the majority of the yield response. Fiscally constrained advanced economies exhibit an eight basis point decline in the expected rate path (significant at the 1% level), while the term premium response is economically negligible and statistically insignificant. In fiscally unconstrained advanced economies, the expected rate path is mildly positive, accompanied by a small but marginally significant term premium increase of 2.5 basis points. The overall yield dynamics in advanced economies after large natural disasters thus predominantly reflect a repricing of the anticipated path of monetary policy which interacts with the fiscal position of the sovereign.

In emerging markets, the term premium plays a substantially larger role. Lower-rated economies show a significant seven basis point decrease in the term premium, accounting for roughly half of the total yield decline, while the expected rate path is negative but imprecisely estimated. Higher-rated economies exhibit significant increases in both components: the expected rate path rises by 7.4 basis points and the term premium by 6.2 basis points, contributing roughly equally to the total yield increase. The stronger term premium signal in emerging markets is consistent with inflation risk and debt issuance expectations playing a more prominent role in yield dynamics for this group.

These results establish that the yield curve response to extreme weather events is heterogeneous in sign, magnitude, and composition across countries with different fiscal regimes. Strikingly, countries with a weaker fiscal position see their yields decline. In advanced economies with a high debt-to-GDP ratio, this decline is entirely driven by a downward revision of monetary policy rate expectations. Among emerging market economies, the decline of yields in poorly-rated countries reflects both a decline in policy rate expectations as well as term premiums. In contrast, emerging market economies with a higher credit rating experience rising yields, which reflect both a steeper expected rate path and an increase of term premiums.

The evidence of state-dependence of sovereign yields to extreme weather events extends Anyfantaki et al. (2025) who document debt-regime heterogeneity in 10-year headline yield responses at the annual frequency. We show that the heterogeneity arises at a much higher frequency, holds at the short end of the curve, and operates through distinct expected-rate-path and term-premium channels.

Table 1: Sovereign yield curve responses to extreme weather events at 50-day horizon

	Advanced Economies		*Emerging Markets*		*Difference*	
	High debt	Low debt	Low rating	High rating	AE	EM
	(1)	(2)	(3)	(4)	(2)−(1)	(4)−(3)
Panel A: 2-year sovereign yield						
Weather shock	-9.56***	5.95	-15.73**	12.71***	15.51**	28.44***
	(3.59)	(4.20)	(6.83)	(4.56)	(7.03)	(10.09)
Panel B: Expected short rate path						
Weather shock	-8.18***	2.97	-9.94	7.40**	11.15**	17.34**
	(2.74)	(3.19)	(6.19)	(3.18)	(5.07)	(8.52)
Panel C: Term premium						
Weather shock	-0.61	2.50*	-7.07**	6.20***	3.12	13.27***
	(1.37)	(1.50)	(2.77)	(1.88)	(2.46)	(4.04)
Observations	162,584	162,584	84,929	84,929		
Countries	18	18	12	12		
Macro controls	✓	✓	✓	✓		
Country FE	✓	✓	✓	✓		
Year FE	✓	✓	✓	✓		

The table reports the impulse response of the 2-year sovereign yield and its decomposition into average expected policy rates and term premium to extreme weather events at the 50-day horizon in basis points, estimated via state-dependent local projections with a smooth transition function $F(z_{t-1})$ following Auerbach and Gorodnichenko (2012). For advanced economies the state variable z_{t-1} is net debt-to-GDP; for emerging markets it is the World Bank sovereign rating index. The natural disaster is an indicator for events in the top quartile of human impact within country, following CRED (2025). All specifications include country and year fixed effects and three lags of the dependent variable. Standard errors in parentheses are Driscoll-Kraay robust to cross-sectional dependence and heteroskedasticity. The difference columns (AE: (2)−(1), EM: (4)−(3)) report the estimated difference in responses across fiscal states with associated standard errors and significance tests. Yield and yield components are measured in basis points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4.2 The Fiscal Space Mechanism

The heterogeneous impulse responses reported in the previous section are consistent with the following mechanism: while countries with a stronger fiscal position expand their government expenditures to stem the economic fallout from a severe weather event, those that have constrained fiscal capacity predominantly rely on monetary accommodation. If this were indeed the case, a natural question is whether fiscally less constrained sovereigns indeed increase government spending following extreme weather events while constrained governments do not. Figure 2 sheds light on this question by showing state-dependent impulse responses of general government expenditure as a share of GDP, using annual data from the IMF World Economic Outlook and following Equation (5).

The contrast across fiscal regimes is stark. In high-debt advanced economies (Panel A), government expenditure as a share of GDP is slightly negative, drifting to approximately -0.5 percentage points two years after the disaster. This is consistent with binding fiscal constraints which prevent or reduce the room for fiscal support. In sharp contrast, low-debt advanced economies (shown in Panel B), see government expenditures significantly rise by about 0.5 percentage points of GDP in the first year and peak at roughly 1.0 percentage points after three years.

The emerging market results mirror this pattern. Lower-rated economies (Panel C) show nearly unchanged government expenditures following disasters, consistent with the absence of fiscal capacity to mount a reconstruction response. In sharp contrast, however, higher-rated economies (Panel D) exhibit a strong and persistent fiscal response: government expenditures rise steadily from approximately 0.2 percentage points on impact to about one percentage point four years after the disaster, with the lower confidence band clearly above zero from year two onward. These results provide evidence that fiscal space is not merely a correlate of the yield response but a direct determinant of the policy channel through which extreme weather events propagate. Unconstrained sovereigns increase spending, constrained governments do not.

The asymmetric fiscal response that we uncover extends Noy and Nualsri (2011), who document counter-cyclical fiscal policy after disasters in developed economies and pro-cyclical responses in developing ones. Our results complement their findings by showing that the relevant state variable is fiscal capacity rather than national income. The remainder of this section studies how this fiscal divergence transmits into yield curve components and inflation dynamics.

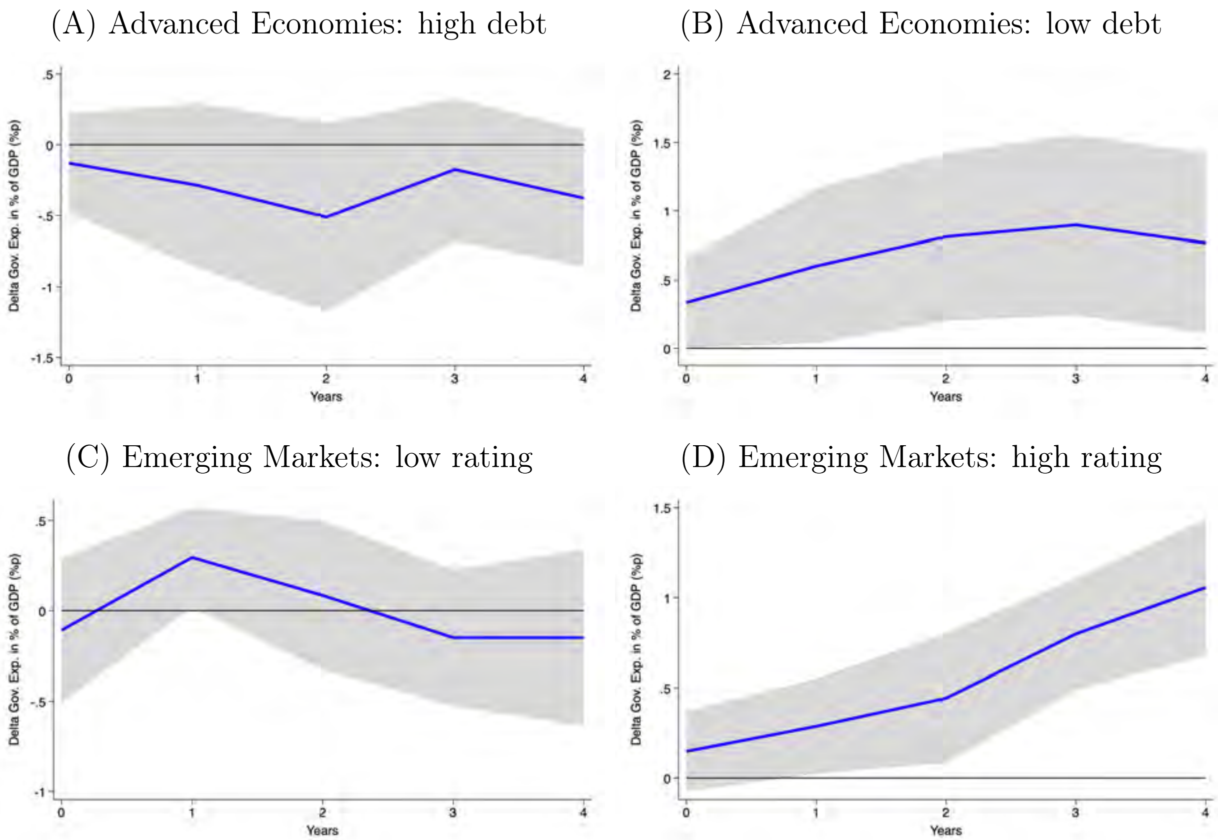


Figure 2: **State-dependent impulse responses of government expenditures in % of GDP.** The figure shows impulse response functions of the annual government expenditures in % of GDP to the largest quartile of extreme weather events, as specified in Equation (5). The left panel shows the impulse responses of countries in a fiscally constrained position and the right panel in a fiscally unconstrained position. The data observation frequency is annual, the sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

4.3 Advanced Economies

This section summarizes the results from state-dependent local projections for advanced economies. We start with a comparison of how policy path expectations and term premiums respond to extreme weather events and then turn to an analysis of the inflation response to these shocks.

4.3.1 Yield Curve Decomposition

Figure 3 presents the state-dependent impulse responses of the two-year sovereign bond yield and its components for advanced economies over the 100-day horizon.

High-debt economies. In fiscally constrained advanced economies, the two-year yield declines significantly following extreme weather events, with point estimates reaching approximately negative ten basis points around 40–60 days after the disaster before gradually reverting toward zero (Panel A). The yield decomposition reveals that this decline is entirely driven by the expected short rate path (Panel C), which declines slowly to a statistically significant -10 basis points around 50 days after the event. Instead, the term premium response (Panel E) is economically negligible and statistically insignificant, fluctuating around zero.

The yield decomposition is thus consistent with markets pricing in anticipated monetary policy easing in response to extreme weather events. In the absence of fiscal stimulus, confirmed by the flat government expenditure response in Figure 2 above, market participants expect the central bank to accommodate the negative fallout of the weather shock. The near-zero term premium response, in turn, suggests that market participants do not expect additional debt issuance or increased inflation risks.

Low-debt economies. In fiscally unconstrained advanced economies, the two-year yield response is positive and persistent, with point estimates rising by five to ten basis points for about two months after the event (Panel B). The decomposition shows that the rise in yields is driven by both components. The expected policy rate path increases by approximately five to eight basis points (Panel D), consistent with markets anticipating a tightening of monetary policy in response to inflationary pressures arising from increased fiscal spending. At the same time, the term premium also rises gradually (Panel F), consistent with markets pricing increased sovereign debt issuance and higher inflation risk.

Both yield components thus move in the same direction in each state. That said, for ad-

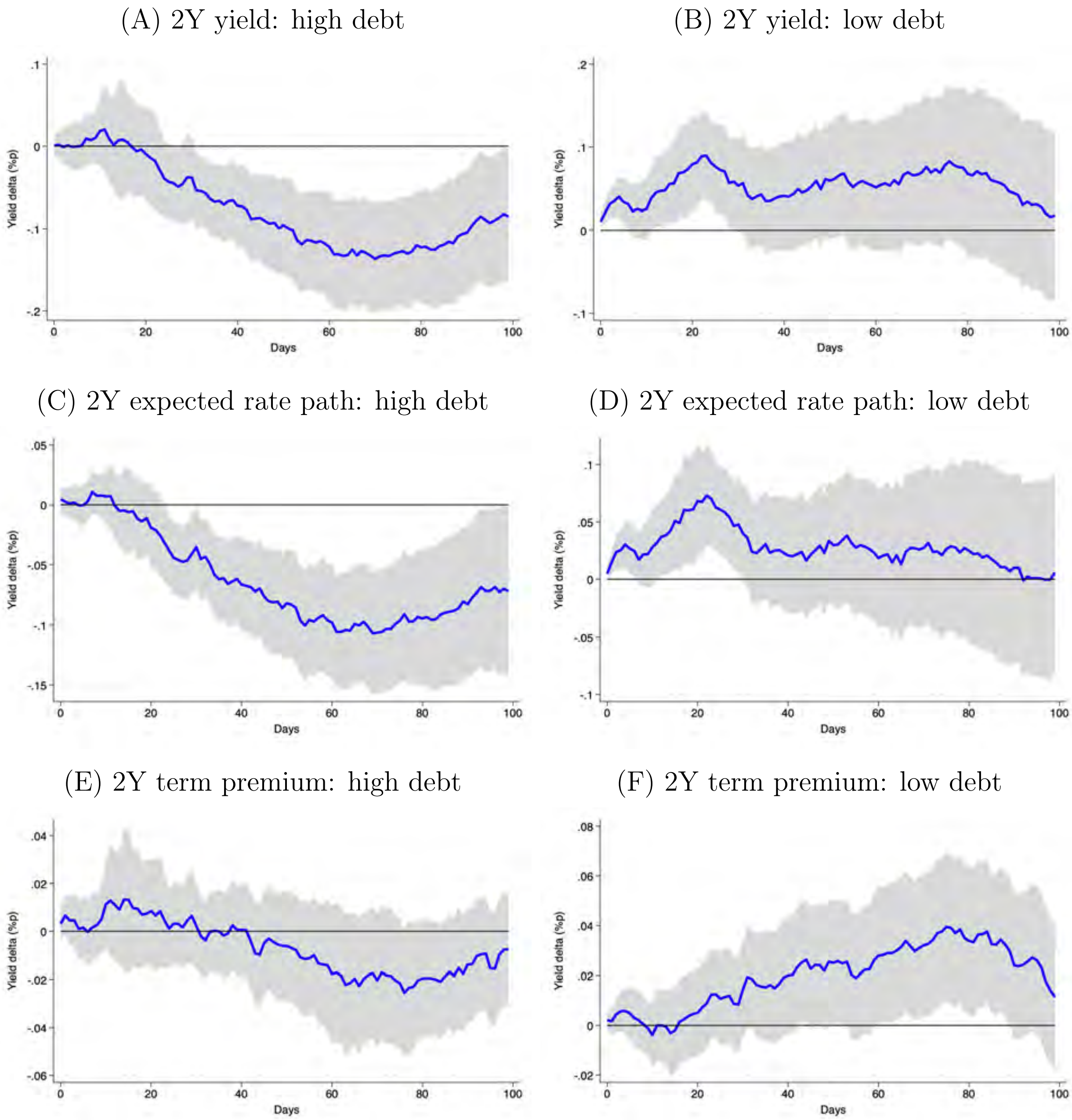


Figure 3: **State-dependent panel impulse responses of advanced economies (daily).** The figure shows impulse response functions of the daily two-year yield and its expected short rate and term premium components to the largest 25% of extreme weather events, as specified in Equation (3). The left panel shows the impulse responses of countries in a high debt-to-GDP state and the right panel in low debt-to-GDP state. The sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

vanced economies the expected rate path is the clear driver of the impulse responses of yields following extreme weather events.

4.3.2 Inflation Dynamics

Figure 4 presents the state-dependent inflation responses to natural disaster shocks. In high-debt advanced economies (Panel A), headline CPI inflation hovers around zero or modestly below throughout the two-year horizon. The response of core CPI (Panel C) confirms this: stripping out food and energy, underlying inflation is essentially flat with a slight negative drift, with point estimates around zero. The absence of an inflationary impulse is consistent with the negative expected rate path documented above—the central bank faces no rise of inflation and can ease policy to cushion the negative real economic effect of disasters.

In striking contrast, in low-debt economies (Panel B), headline CPI rises persistently, and reaches levels about 0.3 percentage points higher one year after the shock. The core CPI response (Panel D) shows a similar positive and persistent pattern, with point estimates stabilizing around 0.2 to 0.3 percentage points. A plausible explanation of this response is the following: in countries where the fiscal space permits additional government spending, this generates demand pressures and higher inflation. In contrast, where fiscal space is absent, no such impulse materializes.

Combined, the evidence thus suggests that extreme weather events lead to increased fiscal spending in countries that have the fiscal space, creating inflationary pressures and markets pricing a steepening expected policy rate path. At the same time, the rise in spending increases the term premium, contributing to a positive overall yield response. In the absence of fiscal space, the chain breaks at the first link, and the dominant effect is monetary accommodation of a disinflationary extreme weather shock. The inflationary dynamics that we document complement the results of Parker (2018), who reports significant but heterogeneous effects following large disasters across countries.

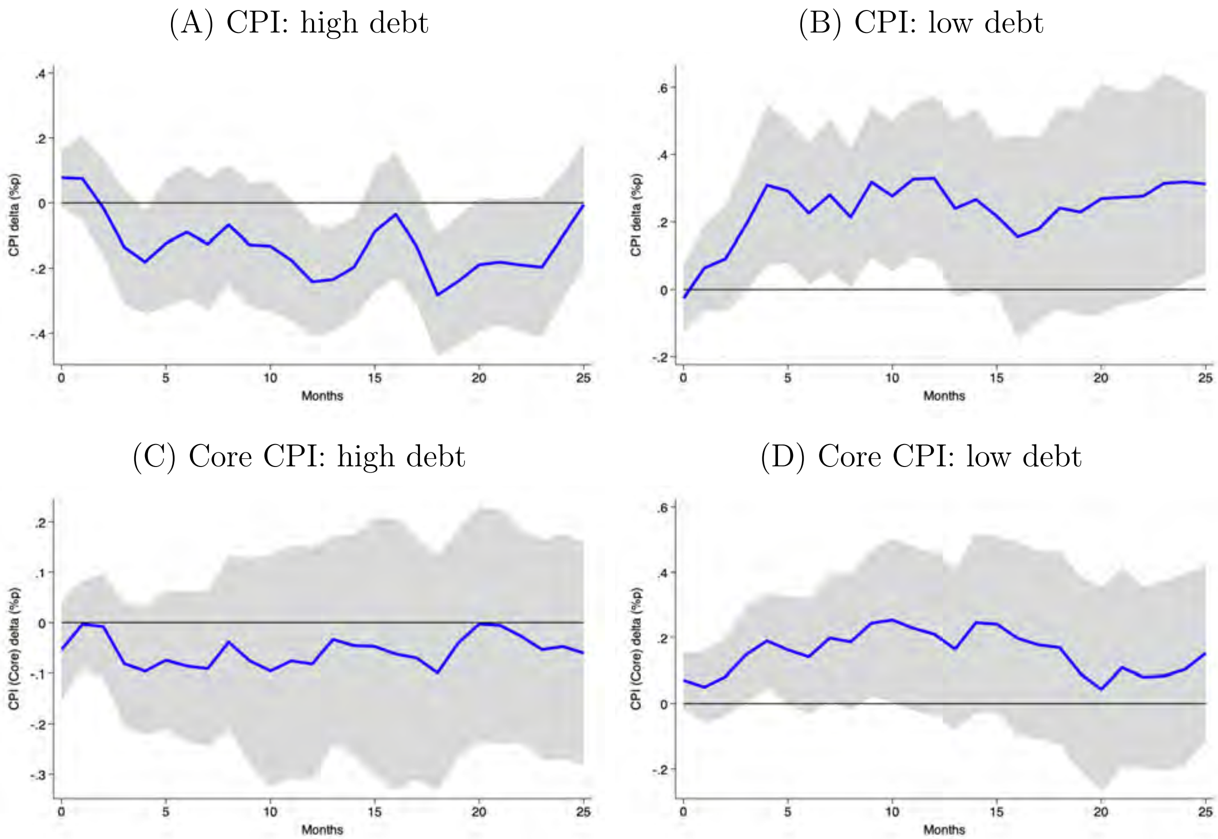


Figure 4: **State-dependent panel impulse responses of advanced economies (CPI).** The figure shows impulse response functions of the monthly headline consumer price index (CPI) and core CPI to the largest 25% of extreme weather events, as specified in Equation (5). The left panel shows the impulse responses of countries in a high debt-to-GDP state and the right panel in low debt-to-GDP state. The sample period follows data availability and is summarized in Table 4. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

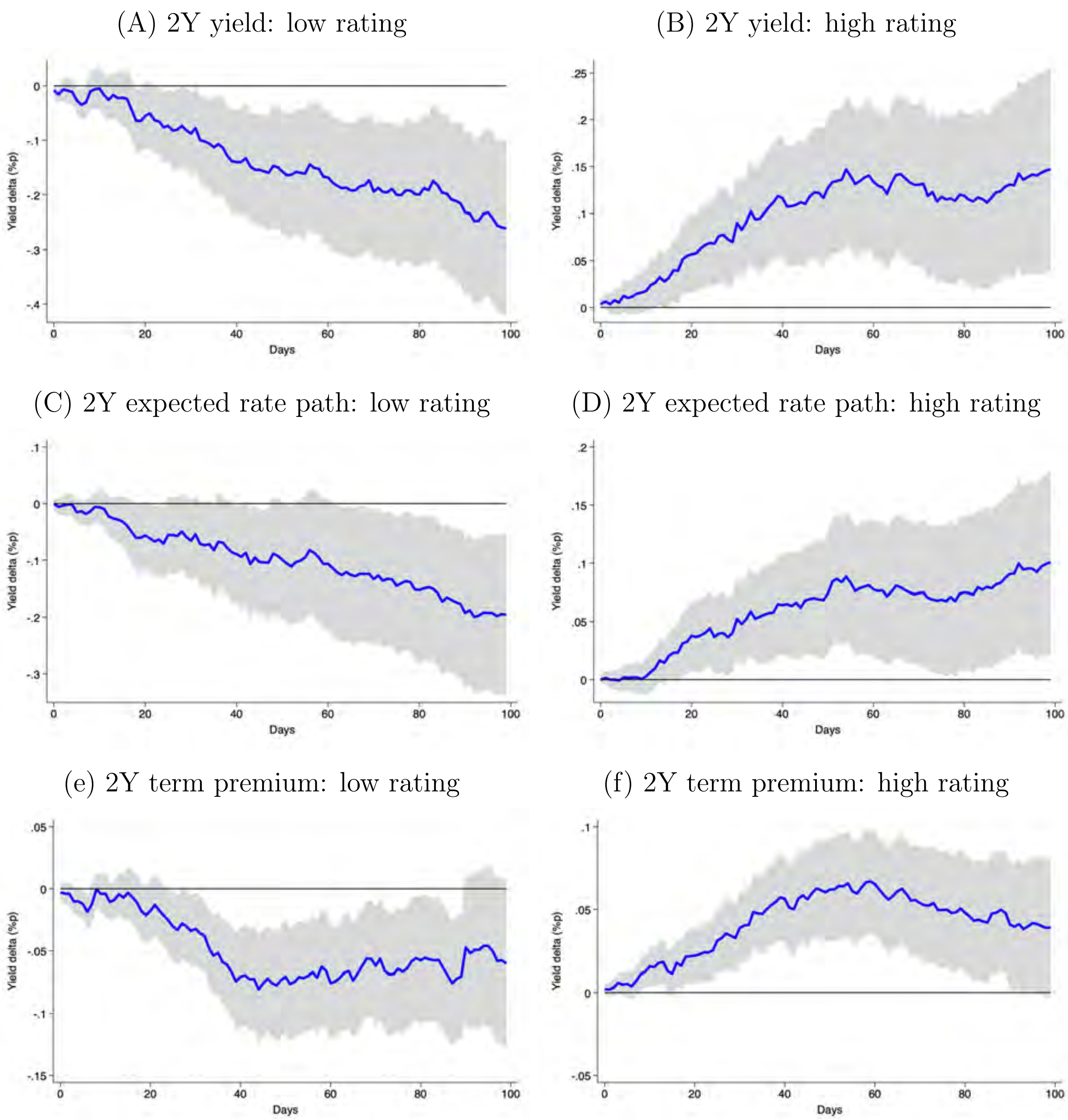


Figure 5: **State-dependent panel impulse responses of emerging markets (Daily).** The figure shows impulse response functions of the daily two-year yield and its expected short rate and term premium components to the largest 25% of extreme weather events, as specified in Equation (3). The left panel shows the impulse responses of countries in a low rating state and the right panel in a high rating state as measured by the World Bank sovereign rating index. The sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

4.4 Emerging Markets

This section summarizes the results from state-dependent local projections for emerging market economies. We first compare how policy path expectations and term premiums respond to extreme weather events and then study the response of inflation to these shocks.

4.4.1 Yield Curve Decomposition

Figure 5 presents the impulse responses for emerging market yields and their components. The yield responses exhibit larger magnitudes and a different decomposition compared to the results for advanced economies, reflecting a greater vulnerability of emerging economies to extreme weather events.

Lower-rated economies. In fiscally constrained emerging market economies, as captured by a low sovereign rating, the two-year yield declines sharply and persistently, with point estimates reaching about minus 25 basis points after 100 days (Panel A). Unlike in the AE high-debt case, where the expected rate path drives the yield compression, both components contribute in emerging markets. The expected rate path declines by approximately 10 to 15 basis points, albeit with wide confidence bands reflecting elevated cross-country heterogeneity (Panel C).

The term premium declines significantly but somewhat delayed (Panel E). This stands in contrast to the evidence for fiscally constrained advanced economies. The absence of fiscal capacity to finance government spending, confirmed by the near-zero government expenditure response in Figure 2, is consistent with market participants not expecting additional sovereign debt issuance. At the same time, as we will show below, the absence of fiscal and monetary accommodation gives rise to a slightly disinflationary inflation response, which is also in line with a lower premium.

Higher-rated economies. In fiscally less constrained emerging markets, the two-year yield rises significantly and persistently by about 10 to 15 basis points and remains elevated throughout the 100-day horizon (Panel B). The decomposition reveals that both the expected rate path and the term premium contribute. The expected rate path rises by about 10 basis points (Panel D), while the term premium increases to a somewhat smaller degree (Panel F).

The positive term premium in higher-rated EMs is consistent with markets anticipating fiscal spending and associated sovereign debt issuance, confirmed by the strong and persistent government expenditure increase documented in Figure 2. The positive expected rate path likely reflects inflationary pressure from the fiscal impulse, as corroborated by the CPI evidence below. The magnitudes in both components are larger than in low-debt advanced economies.[10] While Cevik and Jalles (2022) and Kling et al. (2025) document that climate vulnerability raises sovereign spreads on average, our state-dependent decomposition for local-currency yields reveals that the yield response in better-rated emerging markets reflects both higher expected policy rates and higher term premiums following natural disasters.

4.4.2 Inflation Dynamics

The inflation dynamics in emerging markets reveal a somewhat different transmission channel than in advanced economies, reflecting the greater role of food prices in EM consumption baskets and the sectoral incidence of extreme weather events.

Figure 6 presents the results for headline CPI inflation. In low-rated economies (Panel A), headline inflation declines somewhat over the two years after the disaster, but the coefficients are imprecisely estimated. In contrast, headline CPI inflation rises persistently in higher-rated economies (Panel B). Looking at the components of overall inflation, we see that this increase is mainly driven by food inflation, which rises by up to 2.0 percentage points over the first year after the extreme weather event in higher-rated economies while showing a weakly negative response in lower-rated economies that is not statistically significant.

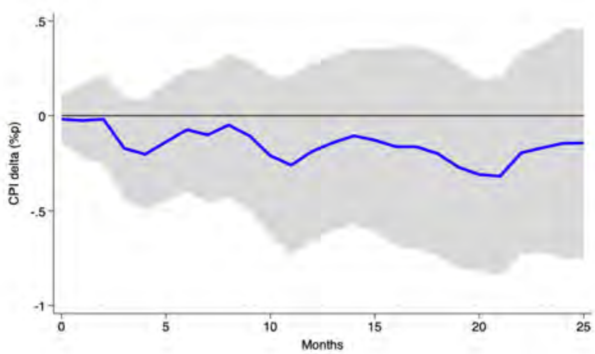
The sharper response of food inflation likely reflects the larger weight of food in emerging market consumption baskets. Extreme weather shocks raise food prices through supply chain disruptions, logistics damage, and the destruction of local farming output. These effects are more likely sustained in countries where disaster relief and reconstruction support aggregate demand. The contrast with advanced economies, where core CPI rather than food prices drives the inflationary response, reflects a structural difference in how extreme weather events transmit into aggregate price dynamics across country groups. Notice further that the food-driven inflationary pressure in poorly rated emerging market economies is consistent with market participants expecting a steeper path of policy rates as documented above.

The dominance of food prices in the emerging markets inflationary response is consistent with Ehlers et al. (2025), who find that food and energy prices drive the short-lived inflationary impact of weather disasters globally, and with Kunawotor et al. (2022) for African economies. The state-dependent shape of the response also resonates with Chavleishvili and Moench

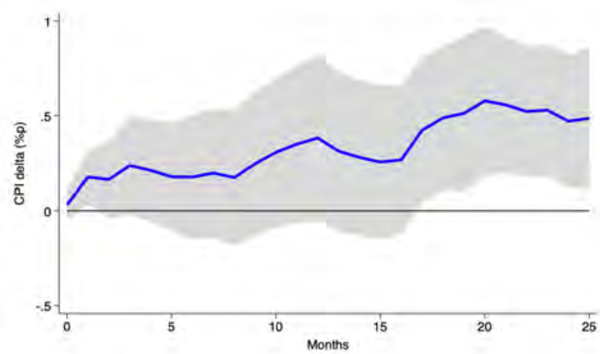
[10] Results are robust across the maturity spectrum. Appendix C reports impulse responses pooled across maturities from 12 months to 10 years, showing a near-parallel shift of the yield curve in both fiscal regimes.

(2025), who document that natural disasters raise upside risk to inflation in a quantile vector autoregressive model estimated for the U.S.

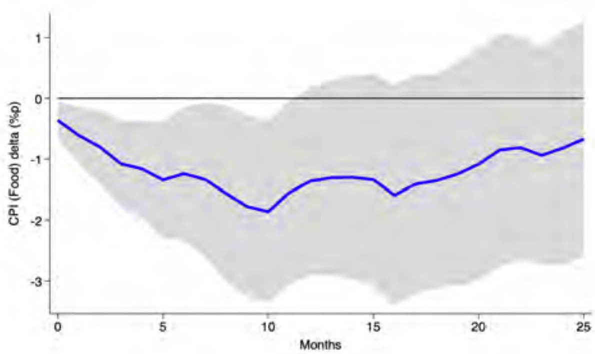
(A) CPI: low rating



(B) CPI: high rating



(C) Food CPI: low rating



(D) Food CPI: high rating

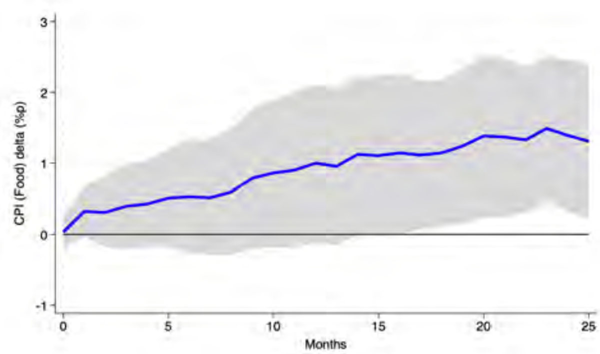


Figure 6: **State-dependent panel impulse responses of emerging markets (CPI).** The figure shows impulse response functions of the monthly headline consumer price index (CPI) and food CPI to the largest 25% of extreme weather events, as specified in Equation (5). The left panel displays the impulse responses of countries with a low sovereign debt rating and the right panel in those with a high rating. The sample period follows data availability and is summarized in Table 4. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

5 Conclusion

The evidence documented in this paper establishes the following results. The unconditional effect of extreme weather events on sovereign yields is approximately zero, a null that masks economically significant heterogeneity. Conditioning on fiscal space reveals that yields decline in fiscally constrained countries and rise in unconstrained ones, a pattern driven by systematically different repricing of monetary policy expectations and inflation risk. Looking at the fiscal response to disasters, we see that government expenditure increases significantly in unconstrained economies and remains flat in constrained ones. This asymmetry is also reflected in post-disaster inflationary dynamics which mainly operate through core inflation in advanced economies and food inflation in emerging markets. The response of expected policy rate and term premium components in sovereign yields mimic these heterogeneous macroeconomic dynamics. In fiscally constrained countries which do not increase government expenditures and where food inflation falls, term premiums and expected policy rates decline. In contrast, in countries that have the fiscal ability to cushion the fallout from the disaster, government expenditures and inflation rise while both expected short rates and term premiums increase.

Our results carry direct implications for policymakers. For governments, the findings imply that increased government expenditures to cushion the economic fallout of disasters – where possible – are associated with higher term premiums on sovereign debt. For central banks, our results indicate that the expansionary fiscal response goes hand in hand with higher inflation and a steeper path of expected policy rates. In sharp contrast, in countries without the fiscal space to mitigate the economic impact of extreme weather events inflation is stable or declines, leading to declining policy rate expectations. Hence, the impact of natural disasters on the term structure of sovereign debt depends on the interactions of fiscal and monetary policy.

Our analysis is subject to a number of limitations. We focus on the short-to-medium-term yield response over a 100-day horizon and do not capture potential long-run repricing of sovereign climate risk that may emerge as physical climate impacts intensify and accumulate. Moreover, the emerging market results, while directionally consistent with the advanced economy findings, are less precisely estimated and should be interpreted with appropriate caution. Finally, our identification captures the financial market response to discrete weather events rather than to the gradual shift in climate risk perceptions that may increasingly shape sovereign bond pricing over longer horizons.

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A Data Availability

Table 2 presents data availability by country as well as the country group composition used in our two panels. The start date is determined by the availability of yield data. The number of events corresponds to all events from that start date until the end of our sample. The number of large events corresponds to the largest 25% of events, as measured by the human impact over the entirety of events in the EM-DAT database for a given country. This allows us to capture longer-term trends in the intensity of extreme weather events despite the comparably short sample period of our yield data. Thus, the ultimate count of large disasters in a given country can be more or less than 25% of the total number depending on the availability of yield curve data for the respective country. Table 3 gives an overview of event type distribution per country within the subset of large events. Table 4 provides an overview of data availability of monthly CPI data by subcomponent and country. The data source in parentheses indicates whether we rely on data from the IMF Data Portal (IMF) or Bloomberg (BBG). Australia and New Zealand are excluded from the monthly inflation impulse response functions as no monthly inflation data is available.

Country	Start date	End date	Events	Large events	Country group
Australia	2000-01-01	2025-12-31	118	30	Advanced economies
Belgium	2000-01-01	2025-12-31	35	9	Advanced economies
Canada	2000-01-01	2025-12-31	76	18	Advanced economies
Denmark	2000-01-01	2025-12-31	9	2	Advanced economies
France	2000-01-01	2025-12-31	115	29	Advanced economies
Germany	2000-01-01	2025-12-31	56	14	Advanced economies
Ireland	2000-01-01	2025-12-31	18	5	Advanced economies
Israel	2005-03-31	2025-12-31	10	3	Advanced economies
Italy	2000-01-01	2025-12-31	77	19	Advanced economies
Netherlands	2000-01-01	2025-12-31	23	6	Advanced economies
New Zealand	2000-01-01	2025-12-31	30	8	Advanced economies
Portugal	2000-01-01	2025-12-31	27	7	Advanced economies
South Korea	2002-11-27	2025-12-31	50	13	Advanced economies
Spain	2000-01-01	2025-12-31	59	15	Advanced economies
Sweden	2000-01-01	2025-12-31	9	2	Advanced economies
Switzerland	2000-01-01	2025-12-31	34	9	Advanced economies
United Kingdom	2000-01-01	2025-12-31	62	16	Advanced economies
United States	2000-01-01	2025-12-31	577	145	Advanced economies
Brazil	2007-03-27	2025-12-31	115	31	Emerging markets
China	2004-04-28	2025-12-31	416	103	Emerging markets
Colombia	2006-04-28	2025-12-31	87	18	Emerging markets
Czech Republic	2000-12-15	2025-12-31	30	8	Emerging markets
Hungary	2001-03-16	2025-12-31	16	4	Emerging markets
India	2000-01-01	2025-12-31	348	87	Emerging markets
Malaysia	2001-09-12	2025-12-31	74	19	Emerging markets
Peru	2006-05-01	2025-12-31	54	13	Emerging markets
Poland	2000-12-15	2025-12-31	43	11	Emerging markets
Romania	2007-01-01	2025-12-31	34	5	Emerging markets
South Africa	2000-01-01	2025-12-31	72	18	Emerging markets
Thailand	2000-01-01	2025-12-31	108	27	Emerging markets

Table 2: Summary of observation days by country.

Country	Storms	Extreme Temperature	Floods	Droughts	Wildfires	Mass movement (dry)	Mass movement (wet)
Australia	12	0	15	0	3	0	0
Belgium	0	3	6	0	0	0	0
Canada	2	0	10	0	6	0	0
Denmark	0	2	0	0	0	0	0
France	6	6	15	0	2	0	0
Germany	3	3	8	0	0	0	0
Ireland	1	0	4	0	0	0	0
Israel	1	0	0	0	2	0	0
Italy	4	4	10	0	1	0	0
Netherlands	0	6	0	0	0	0	0
New Zealand	2	0	6	0	0	0	0
Portugal	1	2	0	0	4	0	0
South Korea	5	0	7	0	2	0	0
Spain	1	3	10	0	1	0	0
Sweden	0	1	1	0	0	0	0
Switzerland	1	2	3	0	0	1	2
United Kingdom	8	1	7	0	0	0	0
United States	66	1	46	0	28	0	1
Brazil	1	0	29	1	0	0	0
China	47	2	57	2	0	0	0
Colombia	3	0	14	0	0	0	4
Czech Republic	3	0	5	0	0	0	0
Hungary	2	0	2	0	0	0	0
India	16	0	71	0	0	0	0
Malaysia	1	0	18	0	0	0	0
Peru	1	2	10	0	0	0	0
Poland	6	1	4	0	0	0	0
Romania	1	1	3	0	0	0	0
South Africa	6	0	11	1	2	0	0
Thailand	4	0	23	0	0	0	0

Table 3: Number of largest events by country and disaster category.

Table 4: CPI Data Availability Start Dates & Source

Country	CPI All Items	CPI Food	CPI Housing	CPI Transportation	Core CPI
Belgium (AE)	1955-01-01 (IMF)	1991-01-01 (IMF)	1998-01-01 (IMF)	1998-01-01 (IMF)	31.01.2000 (BBG)
Brazil (EM)	1979-12-01 (IMF)	2008-08-31 (BBG)	2002-01-01 (IMF)	2002-01-01 (IMF)	n/a
Canada (AE)	1914-01-01 (IMF)	1961-01-01 (IMF)	2014-12-01 (IMF)	2014-12-01 (IMF)	2000-01-31 (BBG)
China (EM)	1993-01-01 (IMF)	2000-01-31 (BBG)	2000-01-31 (BBG)	2000-01-31 (BBG)	2006-01-31 (BBG)
Colombia (EM)	1970-01-01 (IMF)	1995-01-01 (IMF)	2019-12-31 (BBG)	n/a	2010-01-31 (BBG)
Czechia (EM)	1991-01-01 (IMF)	1994-01-01 (IMF)	1995-01-01 (IMF)	1995-01-01 (IMF)	2000-12-31 (BBG)
Denmark (AE)	1967-01-01 (IMF)	1967-01-01 (IMF)	2000-01-01 (IMF)	2000-01-01 (IMF)	2001-01-31 (BBG)
France (AE)	1955-01-01 (IMF)	1970-01-01 (IMF)	1990-01-01 (IMF)	1990-01-01 (IMF)	2000-01-31 (BBG)
Germany (AE)	1955-01-01 (IMF)	1955-01-01 (IMF)	1991-01-01 (IMF)	1991-01-01 (IMF)	2000-01-31 (BBG)
Hungary (EM)	1980-01-01 (IMF)	2001-01-31 (BBG)	2001-01-31 (BBG)	2001-01-31 (BBG)	2001-12-31 (BBG)
India (EM)	1957-01-01 (IMF)	n/a	n/a	n/a	2015-01-31 (BBG)
Ireland (AE)	1955-02-01 (IMF)	1955-02-01 (IMF)	1982-11-01 (IMF)	1982-11-01 (IMF)	2000-01-31 (BBG)
Israel (AE)	1970-01-01 (IMF)	1970-01-01 (IMF)	1985-01-01 (IMF)	1985-01-01 (IMF)	2000-01-31 (BBG)
Italy (AE)	1955-01-01	1957-01-01	1996-01-01	1996-01-01	2002-01-31

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Country	CPI All Items	CPI Food	CPI Housing	CPI Transportation	Core CPI
	(IMF)	(IMF)	(IMF)	(IMF)	(BBG)
Malaysia (EM)	2000-01-01 (IMF)	2000-01-01 (IMF)	2000-01-01 (IMF)	2000-01-01 (IMF)	n/a
Netherlands (AE)	1960-04-01 (IMF)	1960-04-01 (IMF)	1996-01-01 (IMF)	1996-01-01 (IMF)	2000-01-31 (BBG)
Peru (EM)	2010-01-01 (IMF)	2011-01-31 (BBG)	2011-01-31 (BBG)	2011-01-31 (BBG)	2000-01-31 (BBG)
Poland (EM)	1989-01-01 (IMF)	2000-01-31 (BBG)	2000-01-31 (BBG)	2000-01-31 (BBG)	2000-01-31 (BBG)
Portugal (AE)	1955-01-01 (IMF)	1948-01-01 (IMF)	1948-01-01 (IMF)	1948-01-01 (IMF)	2000-01-31 (BBG)
Romania (EM)	2000-01-01 (IMF)	2000-01-31 (BBG)	2000-01-31 (BBG)	2000-01-31 (BBG)	2001-12-31 (BBG)
South Africa (EM)	1957-01-01 (IMF)	2009-01-31 (BBG)	2009-01-31 (BBG)	2009-01-31 (BBG)	n/a
South Korea (AE)	1951-01-01 (IMF)	1985-01-01 (IMF)	1985-01-01 (IMF)	1985-01-01 (IMF)	2000-01-31 (BBG)
Spain (AE)	1954-03-01 (IMF)	2000-01-31 (BBG)	2000-01-31 (BBG)	2000-01-31 (BBG)	2000-01-31 (BBG)
Sweden (AE)	1955-01-01 (IMF)	1955-01-01 (IMF)	1980-01-01 (IMF)	1980-01-01 (IMF)	2000-01-31 (BBG)
Switzerland (AE)	1955-01-01 (IMF)	1955-01-01 (IMF)	1982-12-01 (IMF)	1982-12-01 (IMF)	2005-12-31 (BBG)
Thailand (EM)	2010-01-01 (IMF)	2000-01-31 (BBG)	2000-01-31 (BBG)	2000-01-31 (BBG)	2000-01-31 (BBG)
UK (AE)	1955-01-01 (IMF)	1960-01-01 (IMF)	1988-01-01 (IMF)	1988-01-01 (IMF)	2000-01-31 (BBG)

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Table continued from previous page

Country	CPI All Items	CPI Food	CPI Housing	CPI Transportation	Core CPI
USA (AE)	1955-01-01 (IMF)	2002-12-31 (BBG)	2002-12-31 (BBG)	2002-12-31 (BBG)	2000-01-31 (BBG)

B State Transition Function

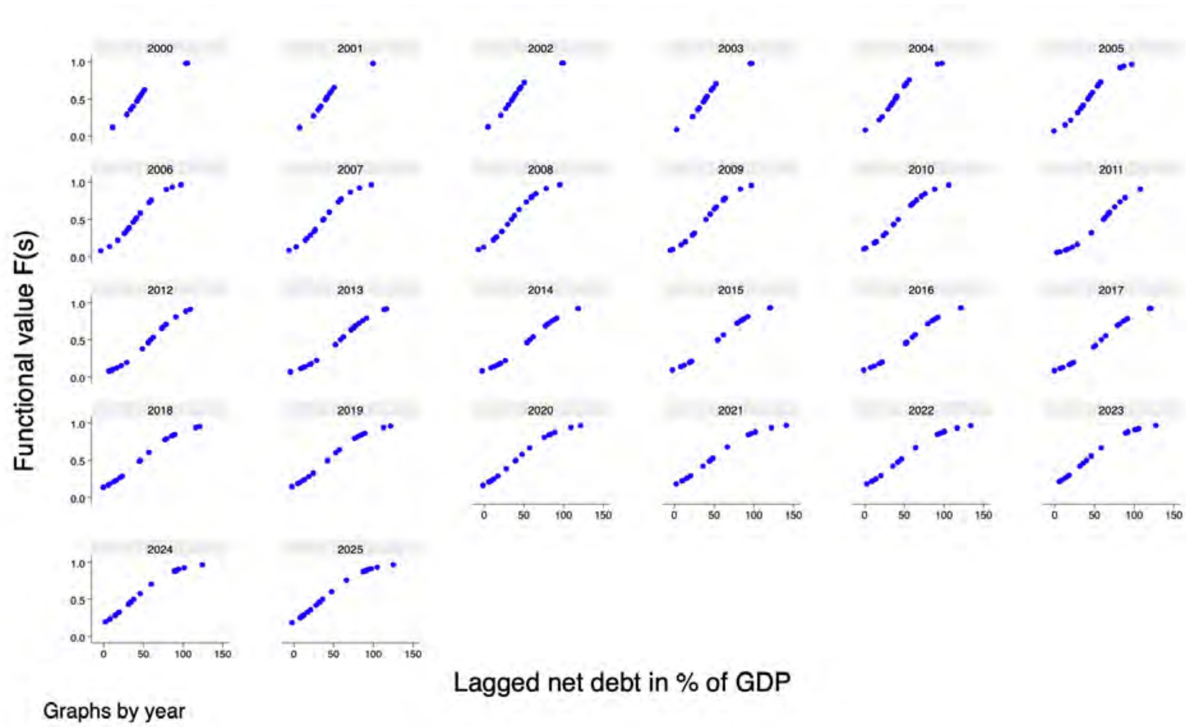


Figure 7: **Smooth transition function** $F(z_{i,t-1})$ **for advanced economies.** The figure plots the logistic transition function following Equation 4 against lagged net debt-to-GDP for each country-year observation, where $\tilde{z}_{i,t-1}$ denotes the standardised state variable and $\theta = 1.5$ following Auerbach and Gorodnichenko (2012). Values of $F(z_{i,t-1})$ close to one indicate the fiscally constrained state; values close to zero indicate the fiscally unconstrained state.

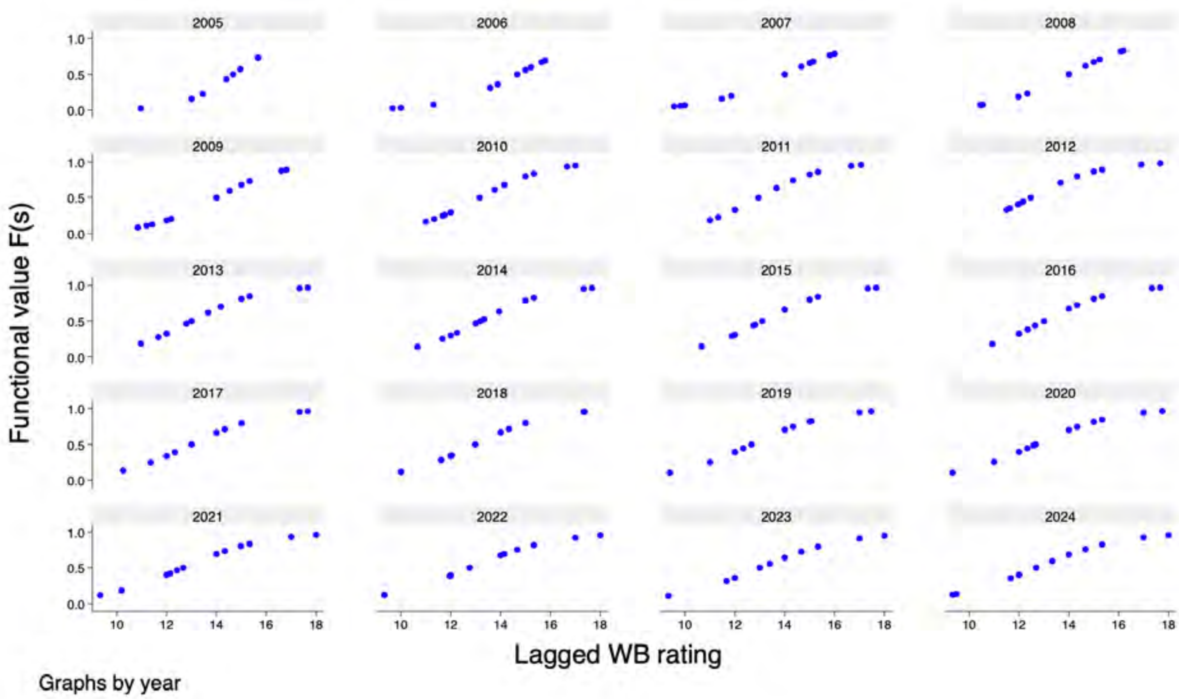


Figure 8: **Smooth transition function** $F(z_{i,t-1})$ **for emerging markets.** The figure plots the logistic transition function following Equation 4 against the lagged World Bank rating index for each country-year observation, where $\tilde{z}_{i,t-1}$ denotes the standardised state variable and $\theta = 1.5$ following Auerbach and Gorodnichenko (2012). Values of $F(z_{i,t-1})$ close to zero indicate the fiscally constrained state; values close to one indicate the fiscally unconstrained state.

Table 5: Sovereign yield curve responses to extreme weather events at 50-day horizon with $\theta = 3$

	Advanced Economies		*Emerging Markets*		*Difference*	
	High debt	Low debt	Low rating	High rating	AE	EM
	(1)	(2)	(3)	(4)	(2)−(1)	(3)−(4)
Panel A: 2-year sovereign yield						
Weather shock	-7.16***	3.27	-11.38*	9.11**	10.44**	20.49**
	(2.70)	(3.31)	(5.87)	(3.80)	(4.95)	(8.09)
Panel B: Expected short rate path						
Weather shock	-6.61***	1.31	-7.30	5.21**	7.92**	12.51*
	(2.24)	(2.56)	(5.34)	(2.56)	(3.72)	(6.84)
Panel C: Term premium						
Weather shock	0.11	1.62	-4.93**	4.43***	1.54	9.36***
	(1.13)	(1.11)	(2.37)	(1.56)	(1.71)	(3.18)
Observations	162,584	162,584	84,929	84,929		
Countries	18	18	12	12		
Macro controls	✓	✓	✓	✓		
Country FE	✓	✓	✓	✓		
Year FE	✓	✓	✓	✓		

The table reports the impulse response of the 2-year sovereign yield and its ACM decomposition to extreme weather eventsat the 50-day horizon in basis points, estimated via state-dependent local projections with a smooth transition function $F(z_{t-1})$ following Auerbach and Gorodnichenko (2012). For advanced economies the state variable z_{t-1} is net debt-to-GDP; for emerging markets it is the World Bank sovereign rating index. The extreme weather events is an indicator for events in the top quartile of human impact within country, following CRED (2025). All specifications include country and year fixed effects and three lags of the dependent variable. Standard errors in parentheses are Driscoll-Kraay robust to cross-sectional dependence and heteroskedasticity. The difference columns (AE: (2)−(1), EM: (4)−(3)) report the estimated difference in responses across fiscal states with associated standard errors and significance tests. Yield and yield components are measured in basis points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Sovereign yield curve responses to extreme weather events at 50-day horizon with $\theta = 5$

	Advanced Economies		*Emerging Markets*		*Difference*	
	High debt	Low debt	Low rating	High rating	AE	EM
	(1)	(2)	(3)	(4)	(2)−(1)	(3)−(4)
Panel A: 2-year sovereign yield						
Weather shock	-6.25**	2.15	-9.40*	7.56***	8.40*	16.96**
	(2.44)	(3.07)	(5.45)	(3.50)	(4.29)	(7.19)
Panel B: Expected short rate path						
Weather shock	-6.07***	0.74	-6.08	4.25*	6.81**	10.34*
	(2.13)	(2.38)	(4.94)	(2.31)	(3.35)	(6.07)
Panel C: Term premium						
Weather shock	0.41	1.09*	-3.98*	3.69**	0.68	7.67***
	(1.09)	(0.97)	(1.44)	(1.44)	(1.51)	(2.81)
Observations	162,584	162,584	84,929	84,929		
Countries	18	18	12	12		
Macro controls	✓	✓	✓	✓		
Country FE	✓	✓	✓	✓		
Year FE	✓	✓	✓	✓		

The table reports the impulse response of the 2-year sovereign yield and its ACM decomposition to extreme weather eventsat the 50-day horizon in basis points, estimated via state-dependent local projections with a smooth transition function $F(z_{t-1})$ following Auerbach and Gorodnichenko (2012). For advanced economies the state variable z_{t-1} is net debt-to-GDP; for emerging markets it is the World Bank sovereign rating index. The extreme weather events is an indicator for events in the top quartile of human impact within country, following CRED (2025). All specifications include country and year fixed effects and three lags of the dependent variable. Standard errors in parentheses are Driscoll-Kraay robust to cross-sectional dependence and heteroskedasticity. The difference columns (AE: (2)−(1), EM: (4)−(3)) report the estimated difference in responses across fiscal states with associated standard errors and significance tests. Yield and yield components are measured in basis points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

C Maturities

(A) 10Y yield: Advanced Economies (B) 10Y yield: Emerging Markets

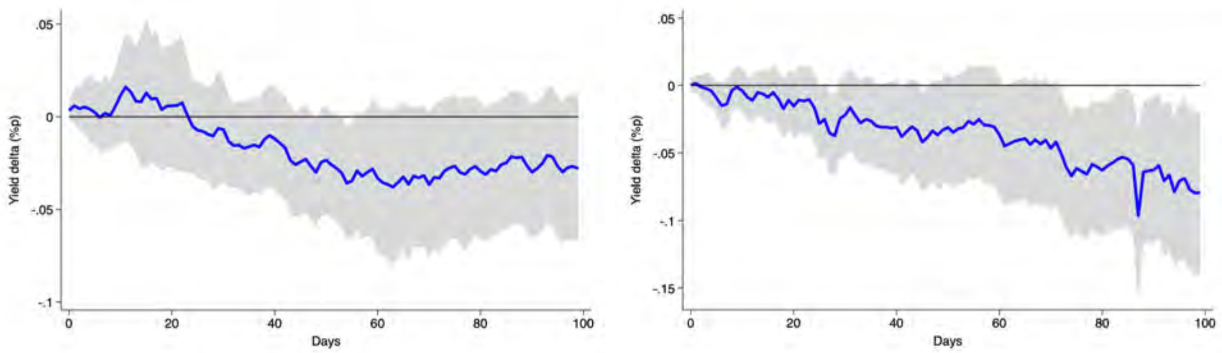


Figure 9: **Impulse responses of 10-year yields to extreme weather events (Daily).** The figure shows impulse response functions of the daily ten-year yield to the largest 25% of extreme weather events, as specified in Equation (2). The left panel shows the impulse responses of countries in the advanced economies group and the right panel in emerging markets group. The sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

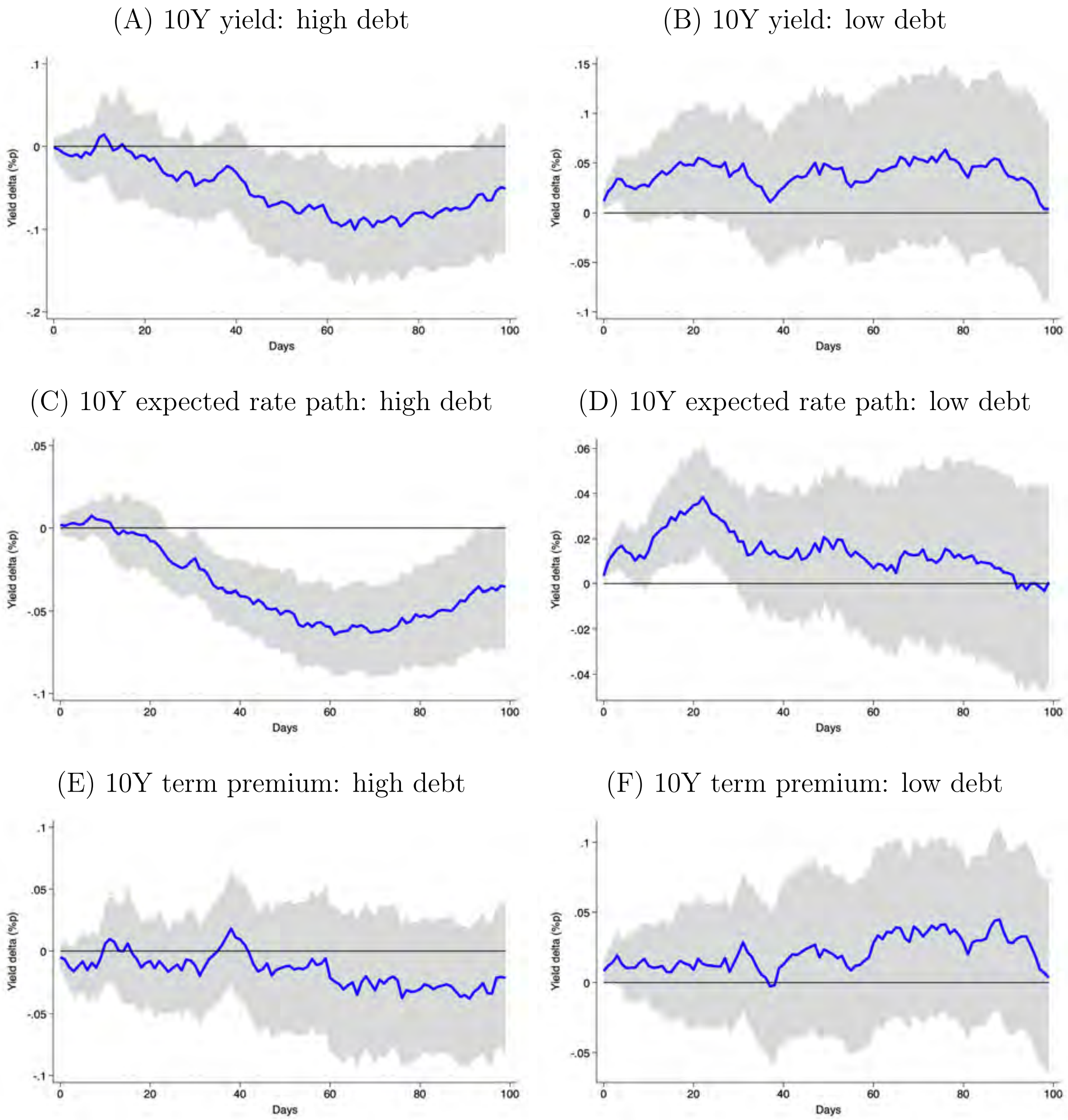
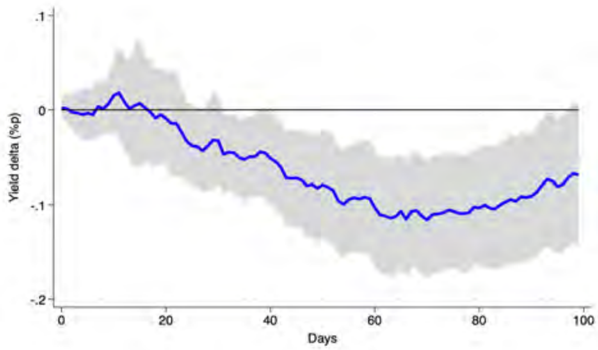
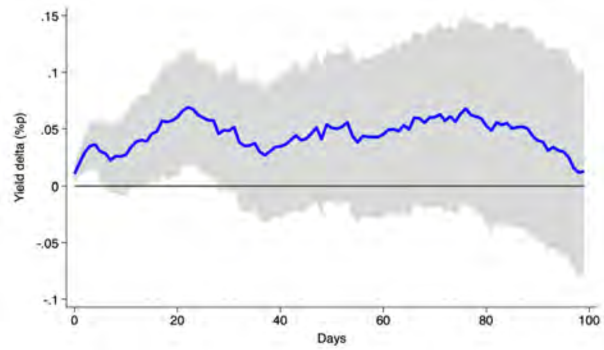


Figure 10: **State-dependent panel impulse responses of advanced economies (daily).** The figure shows impulse response functions of the daily ten-year yield and its expected short rate and term premium components to the largest 25% of extreme weather events, as specified in Equation (3). The left panel shows the impulse responses of countries in a high debt-to-GDP state and the right panel in low debt-to-GDP state. The sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

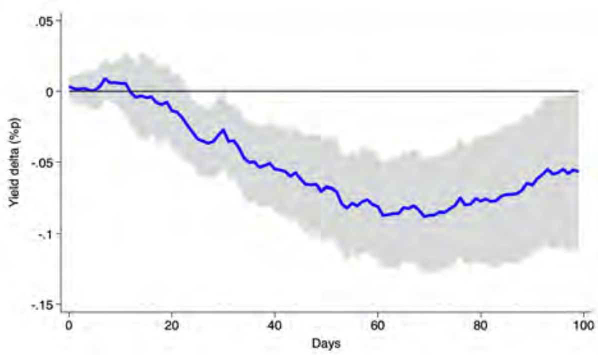
(A) Pooled across maturities: high debt



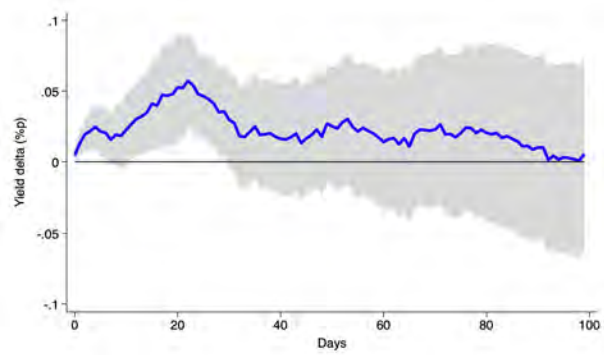
(B) Pooled across maturities: low debt



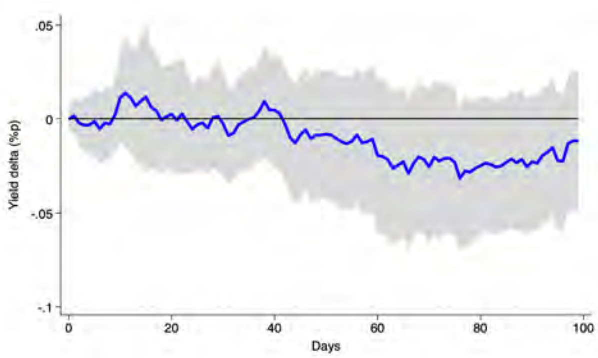
(C) Pooled across maturities: high debt



(D) Pooled across maturities: low debt



(E) Pooled across maturities: high debt



(F) Pooled across maturities: low debt

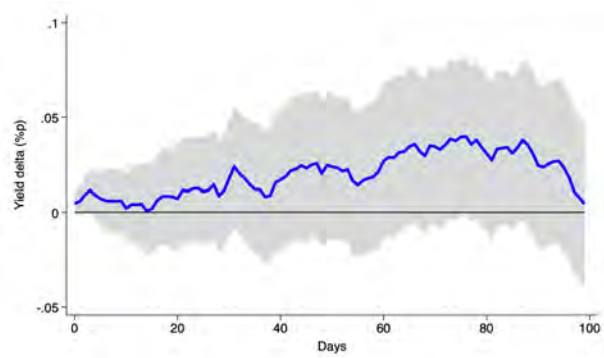


Figure 11: **State-dependent panel impulse responses of advanced economies (daily).** The figure shows impulse response functions for the daily yield and its expected short rate and term premium components to the largest 25% of extreme weather events, as specified in Equation (3) and pooled across 1-year, 2-year, 5-year, 7-year and 10-year maturities. The left panel shows the impulse responses of countries in a high debt-to-GDP state and the right panel in low debt-to-GDP state. The sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

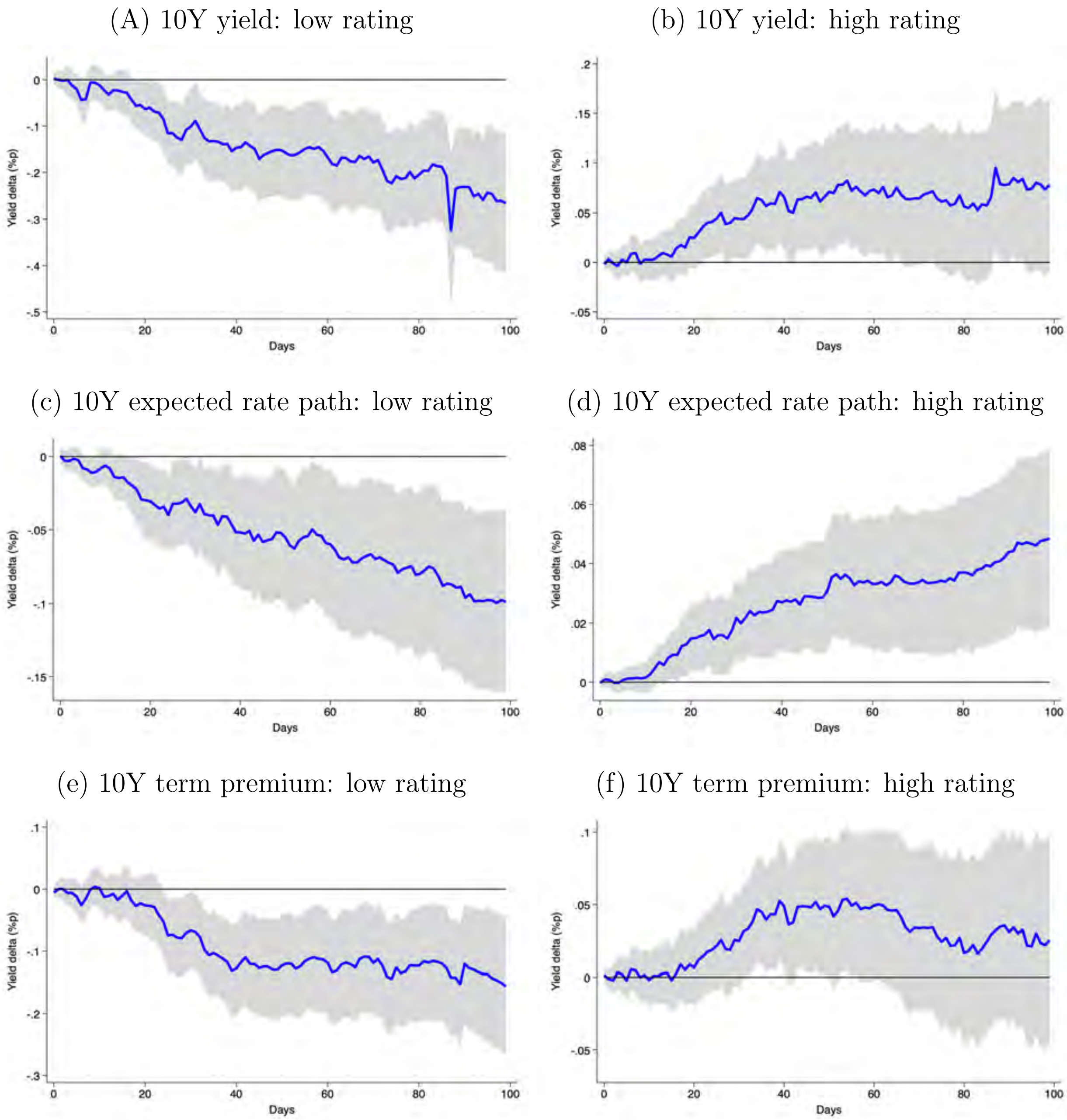
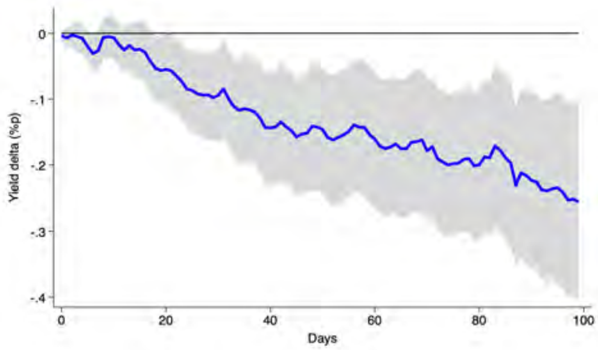
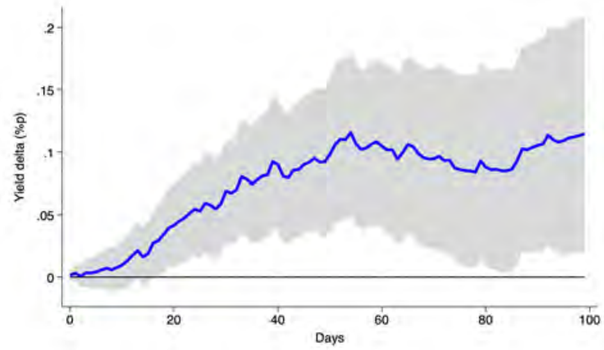


Figure 12: **State-dependent panel impulse responses of emerging markets (Daily).** The figure shows impulse response functions of the daily ten-year yield and its expected short rate and term premium components to the largest 25% of extreme weather events, as specified in Equation (3). The left panel shows the impulse responses of countries in a low rating state and the right panel in a high rating state as measured by the World Bank sovereign rating index. The sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

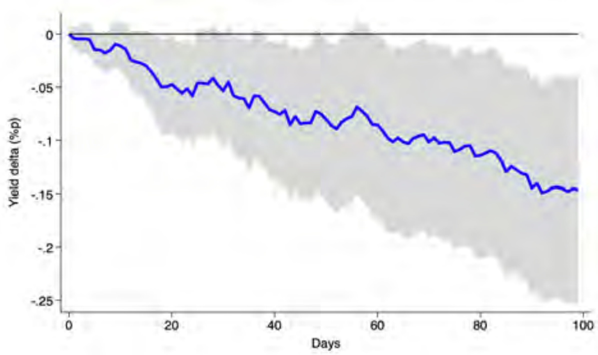
(A) Pooled across maturities: low rating



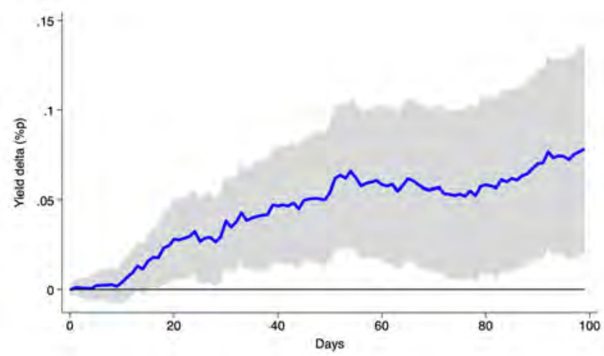
(B) Pooled across maturities: high rating



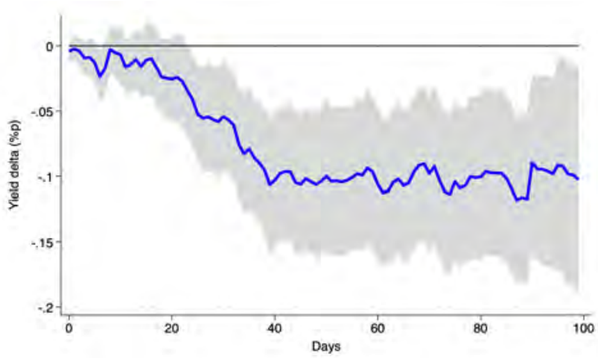
(C) Pooled across maturities: low rating



(D) Pooled across maturities: high rating



(E) Pooled across maturities: low rating



(F) Pooled across maturities: high rating

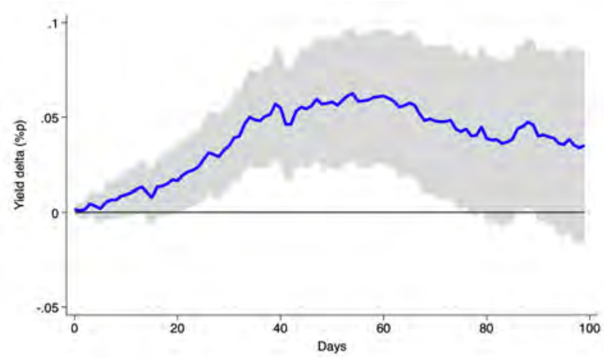


Figure 13: **State-dependent panel impulse responses of emerging markets (Daily).** The figure shows impulse response functions for the daily yield and its expected short rate and term premium components to the largest 25% of extreme weather events, as specified in Equation (3) and pooled across 1-year, 2-year, 5-year, 7-year and 10-year maturities. The left panel shows the impulse responses of countries in alow rating state and the right panel in high rating state. The sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.

D Alternative Severity Thresholds

Table 7: Sovereign yield curve responses to extreme weather events at 50-day horizon with top 33% largest events

	Advanced Economies		*Emerging Markets*		*Difference*	
	High debt	Low debt	Low rating	High rating	AE	EM
	(1)	(2)	(3)	(4)	(2)−(1)	(3)−(4)
Panel A: 2-year sovereign yield						
Weather shock	-8.88***	6.90*	-13.52**	11.71***	15.78**	25.22***
	(3.09)	(3.86)	(3.51)	(3.51)	(6.28)	(7.19)
Panel B: Expected short rate path						
Weather shock	-7.78***	4.30	-10.05**	7.96***	12.09**	18.01***
	(2.38)	(2.97)	(4.88)	(2.42)	(4.67)	(6.56)
Panel C: Term premium						
Weather shock	-0.31	2.26*	-4.72**	4.48***	2.57	9.20***
	(12.10)	(1.32)	(2.20)	(1.47)	(2.18)	(3.22)
Observations	162,584	162,584	84,929	84,929		
Countries	18	18	12	12		
Macro controls	✓	✓	✓	✓		
Country FE	✓	✓	✓	✓		
Year FE	✓	✓	✓	✓		

The table reports the impulse response of the 2-year sovereign yield and its ACM decomposition to extreme weather eventsat the 50-day horizon in basis points, estimated via state-dependent local projections with a smooth transition function $F(z_{t-1})$ following Auerbach and Gorodnichenko (2012). For advanced economies the state variable z_{t-1} is net debt-to-GDP; for emerging markets it is the World Bank sovereign rating index. The extreme weather events is an indicator for events in the top quartile of human impact within country, following CRED (2025). All specifications include country and year fixed effects and three lags of the dependent variable. Standard errors in parentheses are Driscoll-Kraay robust to cross-sectional dependence and heteroskedasticity. The difference columns (AE: (2)−(1), EM: (4)−(3)) report the estimated difference in responses across fiscal states with associated standard errors and significance tests. Yield and yield components are measured in basis points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

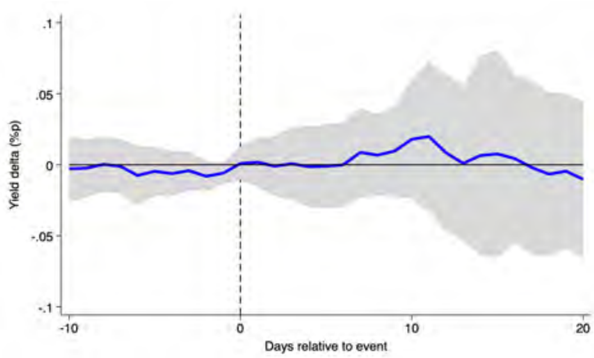
Table 8: Sovereign yield curve responses to extreme weather events at 50-day horizon with top 10% largest events

	Advanced Economies		*Emerging Markets*		*Difference*	
	High debt	Low debt	Low rating	High rating	AE	EM
	(1)	(2)	(3)	(4)	(2)−(1)	(3)−(4)
Panel A: 2-year sovereign yield						
Weather shock	-5.64	4.00	-14.22	8.30	9.60	22.53
	(5.21)	(6.96)	(9.98)	(5.77)	(11.16)	(14.34)
Panel B: Expected short rate path						
Weather shock	-6.94*	4.44	-8.32	6.48	11.38	14.81
	(4.02)	(5.05)	(9.01)	(4.30)	(7.91)	(12.52)
Panel C: Term premium						
Weather shock	1.59	-0.36	-11.39***	4.80**	-1.95	16.19***
	(2.25)	(2.67)	(4.33)	(2.39)	(4.05)	(5.83)
Observations	162,584	162,584	84,929	84,929		
Countries	18	18	12	12		
Macro controls	✓	✓	✓	✓		
Country FE	✓	✓	✓	✓		
Year FE	✓	✓	✓	✓		

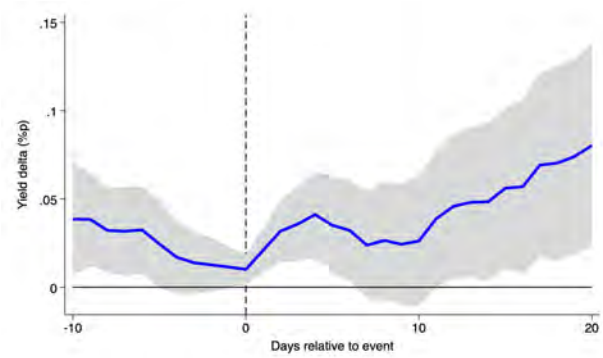
The table reports the impulse response of the 2-year sovereign yield and its ACM decomposition to extreme weather events at the 50-day horizon in basis points, estimated via state-dependent local projections with a smooth transition function $F(z_{t-1})$ following Auerbach and Gorodnichenko (2012). For advanced economies the state variable z_{t-1} is net debt-to-GDP; for emerging markets it is the World Bank sovereign rating index. The extreme weather events is an indicator for events in the top quartile of human impact within country, following CRED (2025). All specifications include country and year fixed effects and three lags of the dependent variable. Standard errors in parentheses are Driscoll-Kraay robust to cross-sectional dependence and heteroskedasticity. The difference columns (AE: (2)−(1), EM: (4)−(3)) report the estimated difference in responses across fiscal states with associated standard errors and significance tests. Yield and yield components are measured in basis points. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

E Pre-trend

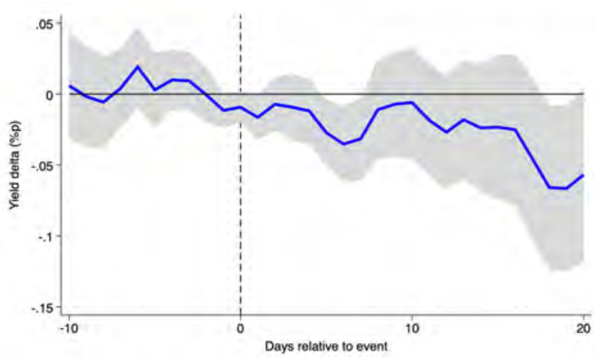
(A) Advanced economies 2Y yield: high debt



(B) Advanced economies 2Y yield: low debt



(C) Emerging markets 2Y yield: low rating



(D) Emerging markets 2Y yield: high rating

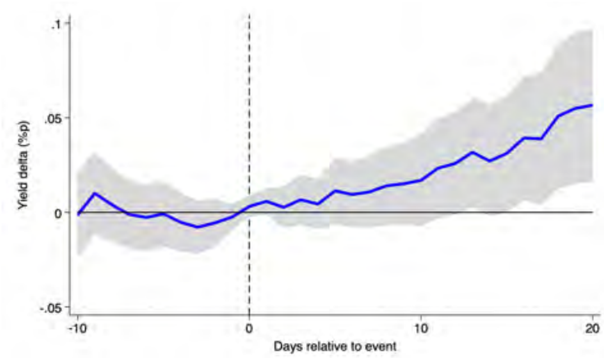


Figure 14: **State-dependent panel impulse responses testing pre-trend (Daily).** The figure shows impulse response functions of the daily 2-year yield from 10 days before to 20 days after the largest 25% of extreme weather events. The specification follows Equation (3) but omits lagged dependent variables to maintain consistency across the pre- and post-event window. Main results with the full control set are reported in Figure 3 (advanced economies) and Figure 5 (emerging markets). The dashed vertical line marks the event date. Panels (A) and (B) show advanced economies in high and low debt-to-GDP states; Panels (C) and (D) show emerging markets in low and high sovereign rating states. The sample period follows data availability and is summarized in Table 2. We use Driscoll-Kraay standard errors and show confidence bands at the 90% level.